

Fundamental of Remote Sensing

Tong Si Son

Email: tong-si.son@usth.edu.vn

Specification: Remote sensing and Applications

Structure Of The Course

- **Time**

  36 hours: Lecturer: 21 h, Practice: 15

- **Contents**

  1. Overview of Remote sensing
  2. Physical basis of Remote sensing
  3. Structure of remote sensing systems
  4. Satellite image characteristics
  5. Satellite image pre-processing
  6. Information extraction on satellite images and accuracy assessment

Section 6. Information extraction on satellite images and accuracy assessment

6.1 Image interpretation

6.2 Land cover/ land use extraction

6.3 Image classification

6.4 Accuracy assessment

6.5 Extracting information from index images

6.6 Practice

6.1 Image interpretation

Image interpretation is a process of identifying what we see on the images and communicate the information obtained from these images to others for evaluating this significance.

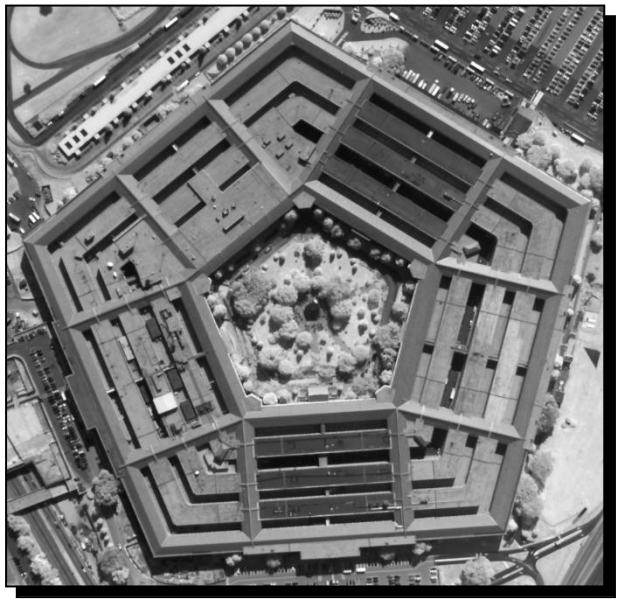
Elements of image interpretation

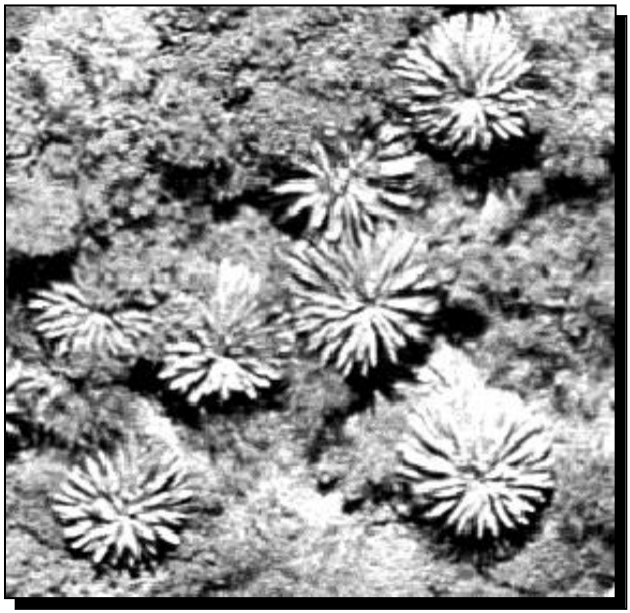
- »Shape
- »Size
- »Tone/Color
- »Texture
- »Pattern
- »Shadow
- »Site
- »Association

Shape

- The outline of a feature
  - Geometric characteristics: linear, curvilinear, circular, elliptical, radial, square, rectangular, triangular, hexagonal, pentagonal, star, amorphous, etc.
  - Important to note that shape depends **on perspective**
  - Overhead perspective, introduces scale effect

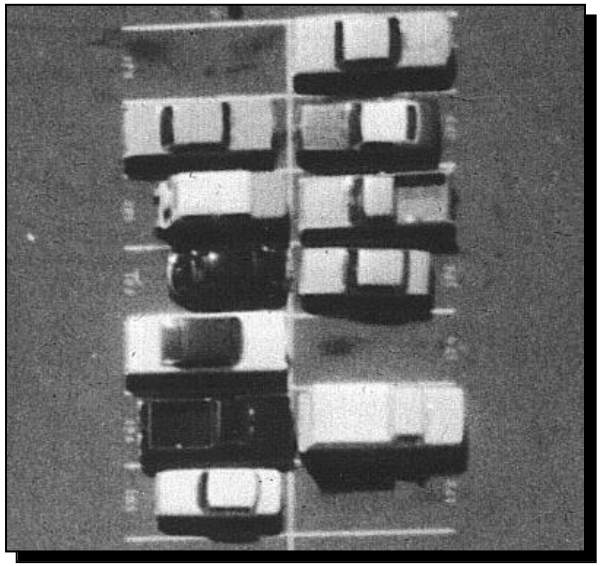




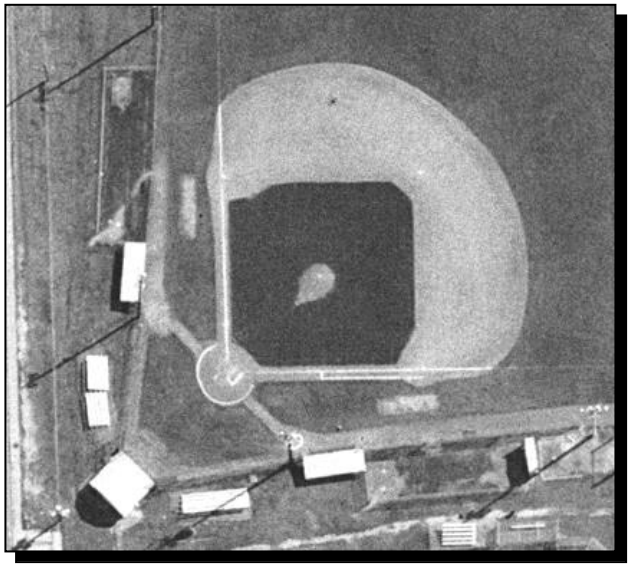


Size

- The dimensions of a feature
- ***Relative size*** determined by comparing the object with familiar nearby features
- ***Absolute size*** refers to the use of the aerial image to derive measurements
- Length, width, perimeter, area (m2) small, medium (intermediate), large



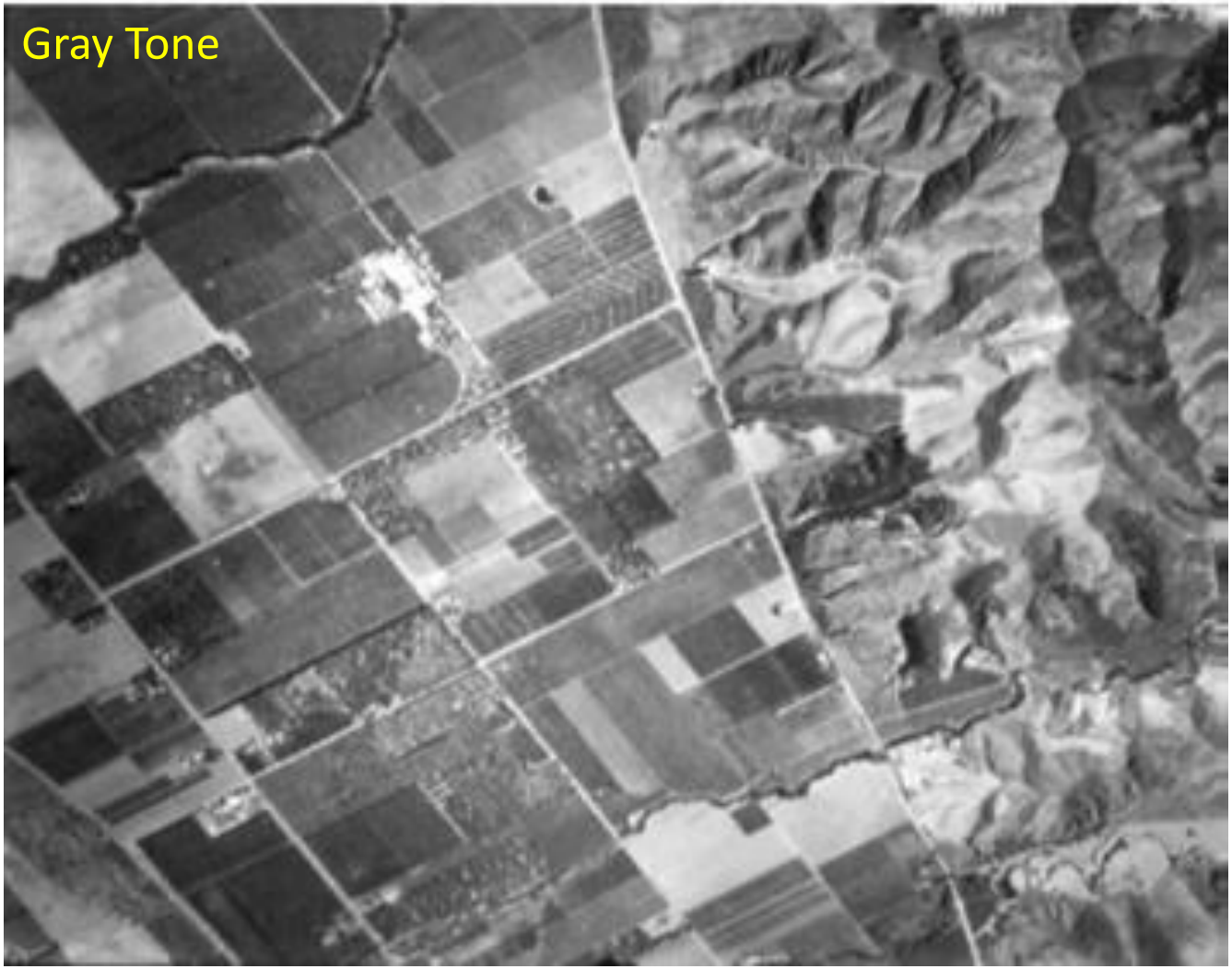




Tone/ Color

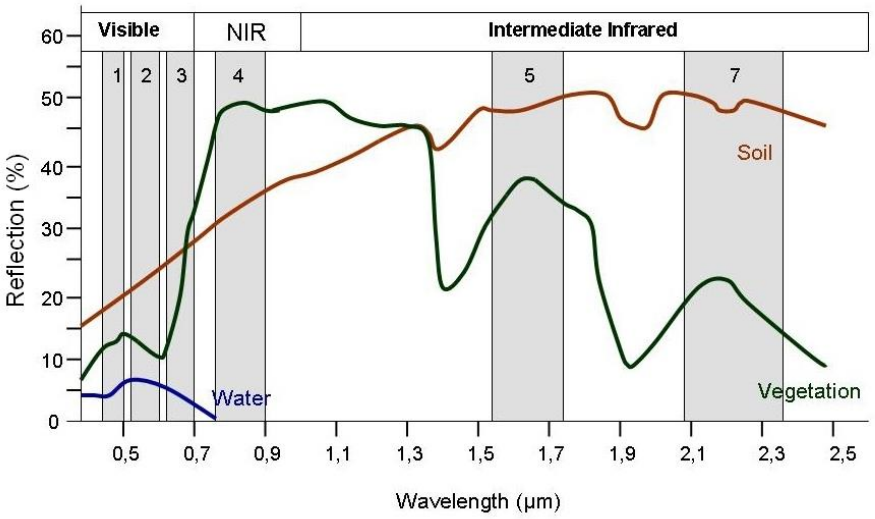
- Recorded EMR:
  - Gray ranging from black to white – tone. (Tone refers to bright levels)
  - Use additive color-combining techniques to create color composite images.
- Refers to the average brightness of an area or, in the case of color imagery, to the dominant color of the region
  - Depends on the nature of the surface in the ankles of observation and illumination.
  - Smooth surfaces behave like ***specular reflectors***, they tend to reflect radiation in a single direction
    - These features may appear bright or dark
  - Rough surfaces behave this ***diffuse reflectors.***
    - Scatter radiation in all directions.
    - A peer is medium gray tones

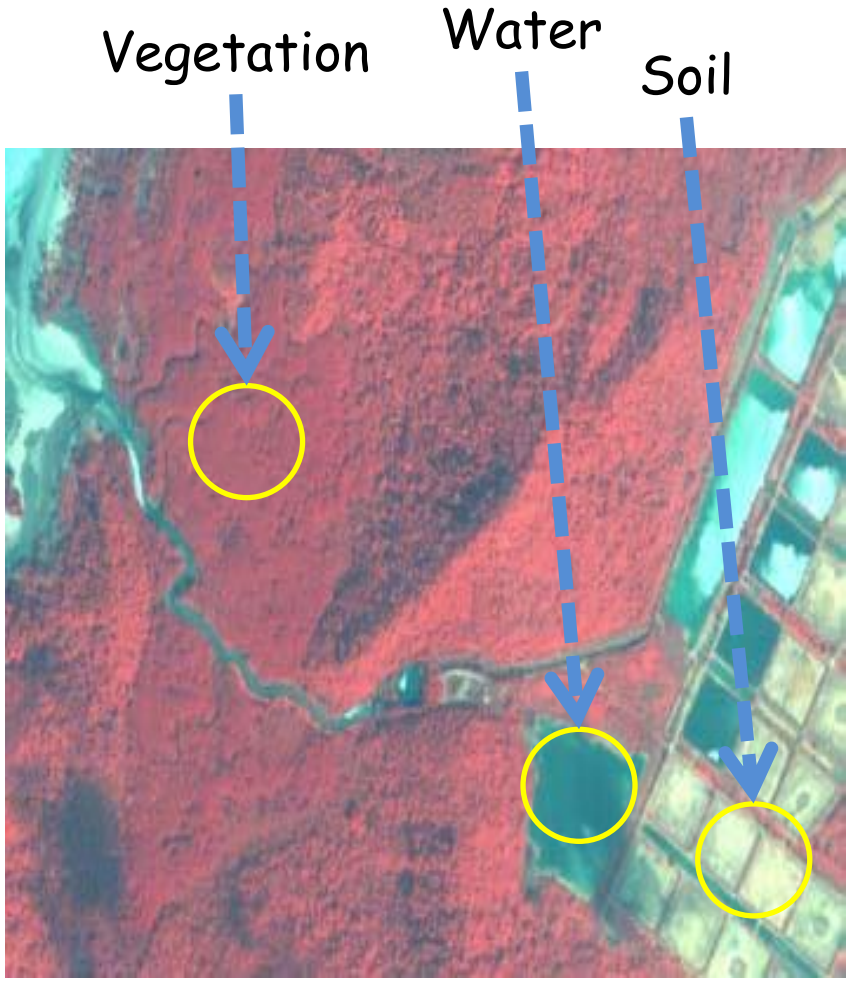
Tone



Color

Spectral reflectance of natural targets

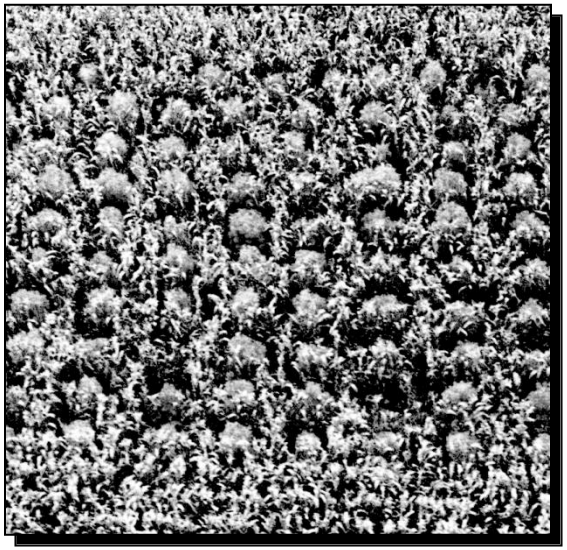




Pseudo color composite:
Near-Infrared, Red light, Green light in RGB combination
What can you see in the image?

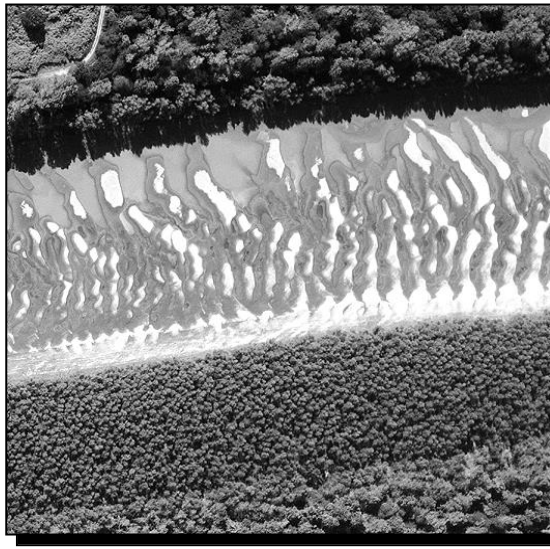
Texture

- Texture is the characteristic placement and arrangement of repetitions of tones or color in an image.
- Refers to the variation in tone over a surface or the apparent roughness of the surface as seen in the photo
- Created by micro shadows in small irregularities in the surface



Perennial trees





Sand

Pattern

- Pattern is the spatial arrangement of objects in the landscape
- Refers to distinctive arrangement of features
  - Orchards have trees plant can rows
  - Laval stream is arranged in rows



Laval



Palm trees

Shadow

- Caused by solar illumination from the side
- Refers to large distinctive shadows that revealed the outline of a future as projected onto a flat surface.
  - Depends on the nature of the object, angle of illumination, perspective, and slope of the ground surface
  - Measure the height (elevation) or depth (bathymetry) of an object or landform



Pyramit



Oil tanks

Site

- Refers to a features position with respect to topography and drainage.
  - Some things occupy a distinctive topographic position because of their function
    - *Sewage treatment facilities at the lowest feasible topographic position.*
    - *Power plants located adjacent to water for cooling*





Association

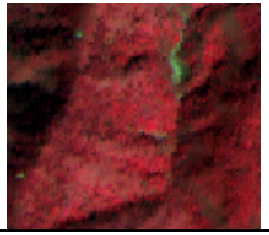
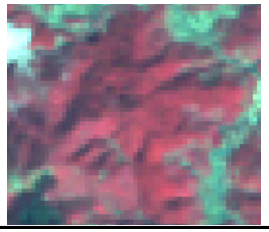
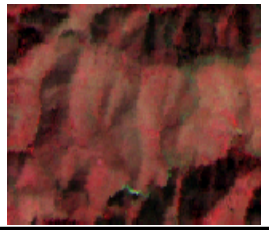
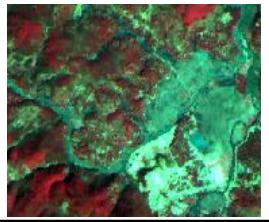
- When you find a certain activity or phenomena, you almost invariably encounter related or associated features or activities.
- Association: related phenomena are usually present
- Association refers to the distinctive spatial interrelationships between features
  - Schools often associated with athletic fields.
  - Large train station often associated with many railways





Key system for classification

Interpretation key = 1 class sample = Σ (Image elements)= Σ (Shape, size, texture, pattern, shallow, tone/color, site)

Land cover system	Image sample	Interpretation keys		Field images
Shrub land		- Light red - Uncertain shape	- Distributed in dow nhill area, buffer zone of the forest, - Often located near by bare land area or sometimes close to residential areas.	
Plantation forest		- Red color - Fine texture	- Located near by residential areas.	
Grass land		- Very light red or purple - Uncertain shape	- Often distribute in the dow nhill area	
Annual crop		It has bright or sea green depends on the grow ing period	-Distribute near by settlement	
Settlement		-Special texture -Bright or dot w hite, clear on image	-Distribute on flat terrain	

6.2 LAND COVER/ LAND USE EXTRACTION

- Land cover is the natural/artificial compositions at the surface

*Land cover data documents **how much** of a region is **covered** by forests, wetlands, impervious surfaces, agriculture, and other land and water types. Water types include wetlands or open water*

*Each group of **above object** is defined as a **class** in land cover map.*

- Land cover classification

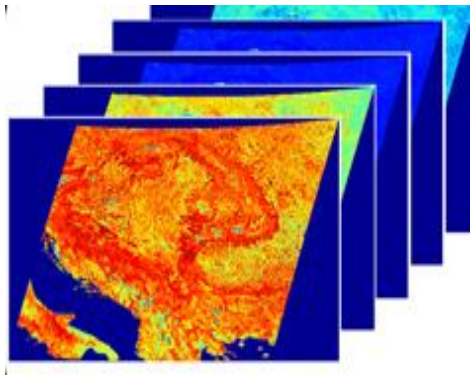
*Classification is the "**ordering or arrangement** of objects into **groups or sets** on the basis of their relationships" (Sokal, 1974). It is an abstract representation of the **situation in the field**, using well-defined diagnostic **criteria**.*

- Classification scheme = land cover classification system
- Images (Satellite images, Aerial photos)
- Interpretation key = class samples
- Methods for classification

www.noaa.gov

How to make Land cover maps?

GLC2000 Global Classes: Forest	
1	Tree Cover, broadleaved, evergreen **(LCCS >15% tree cover, tree height >3m)**
2	Tree Cover, broadleaved, deciduous, closed (or open to closed)
3	Tree Cover, broadleaved, deciduous, open **(LCCS 15% - 40% tree cover, tree height >3m)**
4	Tree Cover, needle-leaved, evergreen
5	Tree Cover, needle-leaved, deciduous
6	Tree Cover, mixed phenology or leaf type
7	Tree Cover, regularly flooded
8	Mosaic: Tree cover / Other natural vegetation



2/ Image data

Class samples, interpretation keys

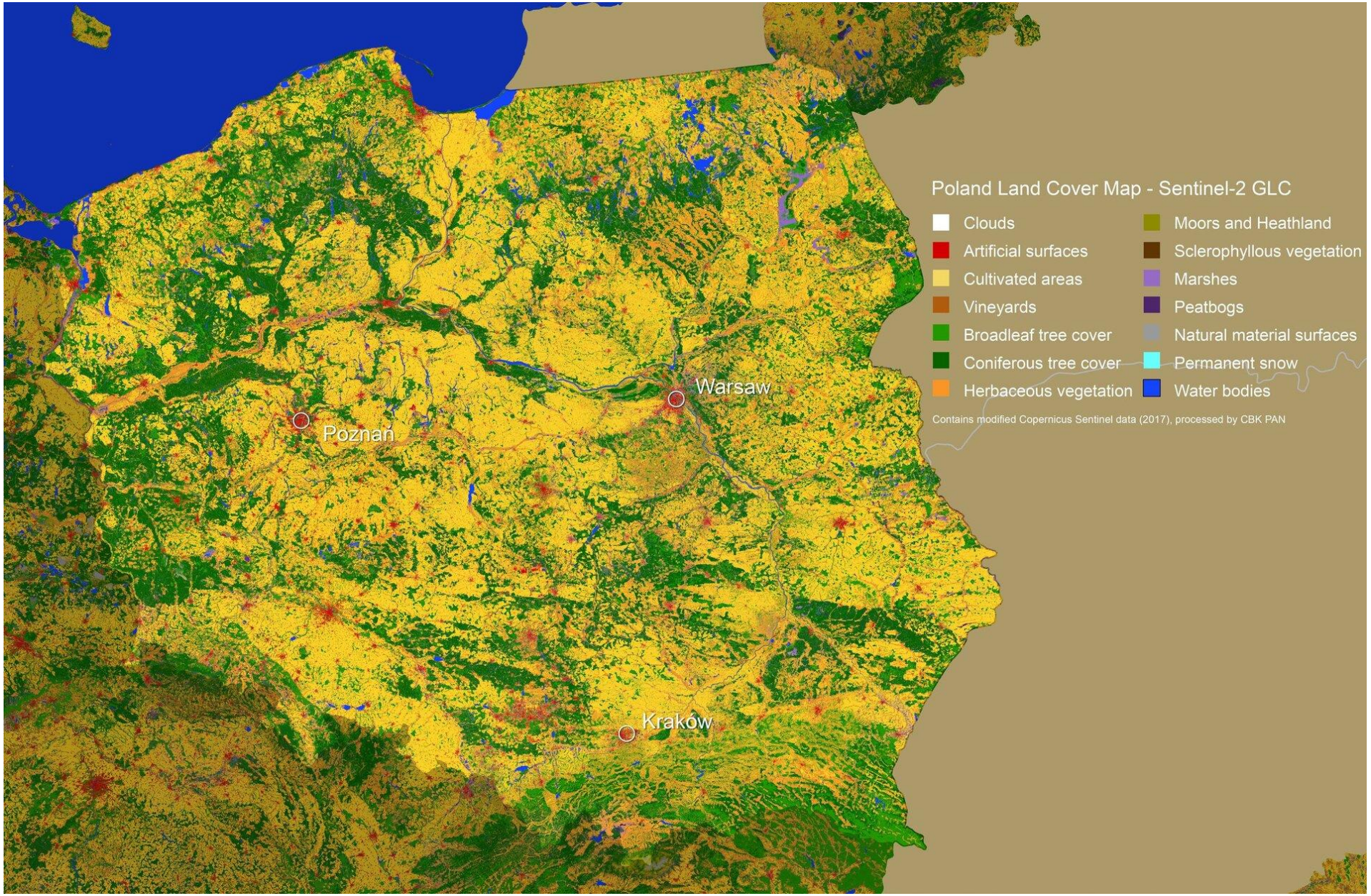
3/ Knowledge strategy

Classification methods
- Digitalization
- Supervised
- Unsupervised
- …

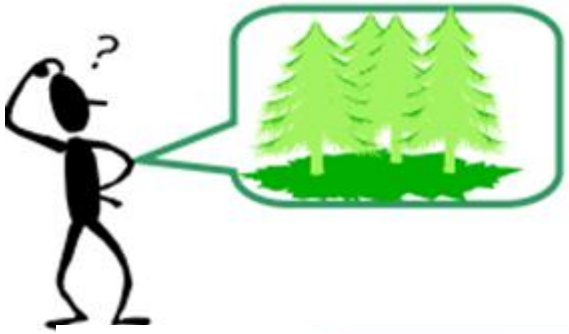
4/ Methods

1/ Define the purposes

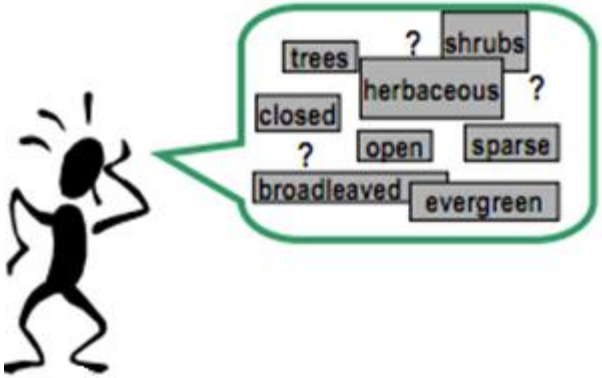



Poland Land Cover Map - Sentinel-2 GLC
Clouds
Artificial surfaces
Cultivated areas
Vineyards
Broadleaf tree cover
Coniferous tree cover
Herbaceous vegetation
Moors and Heathland
Sclerophyllous vegetation
Marshes
Peatbogs
Natural material surfaces
Permanent snow
Water bodies
Contains modified Copernicus Sentinel data (2017), processed by CBK PAN
Warsaw
Poznań
Kraków

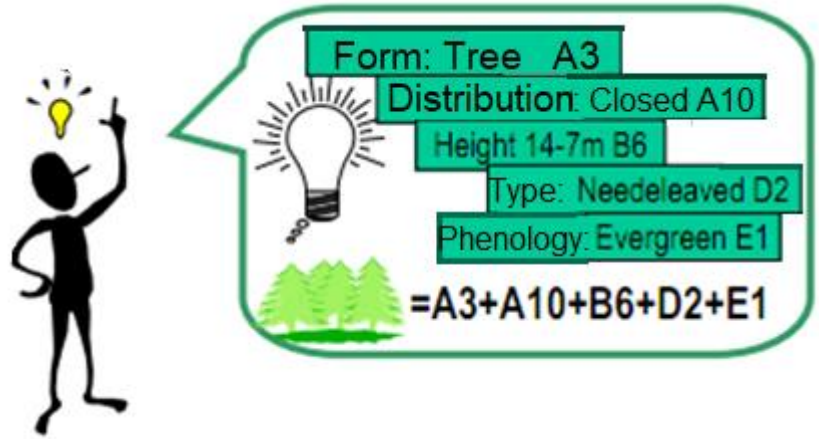
How to create Land Cover classes



Basic concept of a land cover class
(the idea)



Use of Land cover system method
(the language)



Elaboration of the concept in the codified Land cover system language
(the concept expression)

Concept of land cover class

Stepwise selection of classifiers that will generate:

- Criteria Classifiers used
- a string of codes, the so-called Boolean Formula;
- a Standard Class Name; and
- a unique Numerical Code.

Example "Natural and Semi-Natural Terrestrial Vegetation (A12)" :

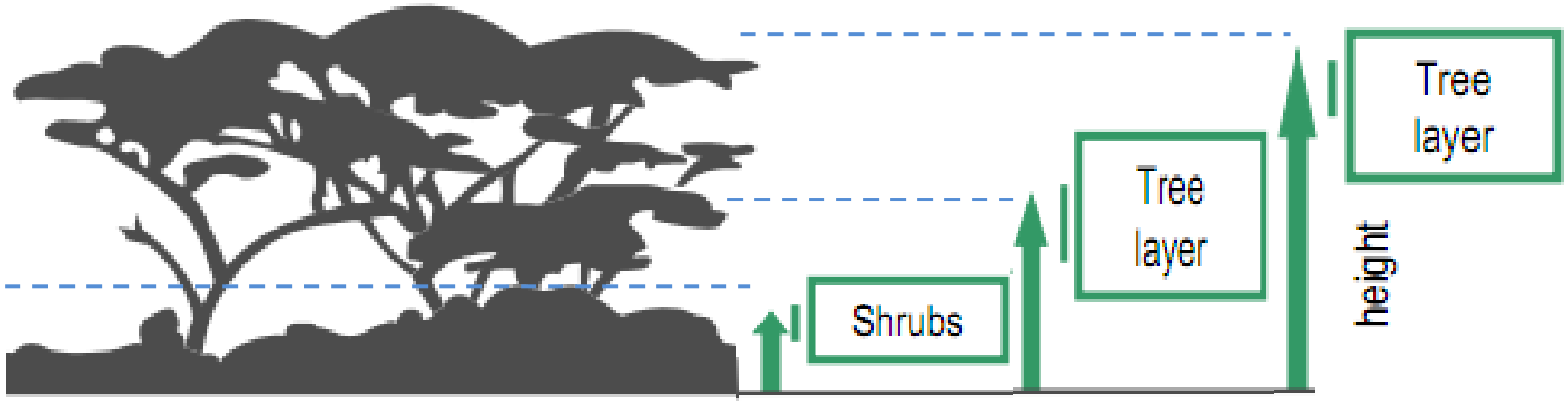
Classifiers Used:	Boolean Formula:	Standard Class Name:	Code:
Life Form & Cover	A3A10	Closed Forest	20005
Height	A3A10B2	High Closed Forest	20006
Spatial Distribution	A3A10B2C1	Continuous Closed Forest	20007
Leaf Type	A3A10B2C1D1	Broadleaved Closed Forest	20095
Leaf Phenology	A3A10B2C1D1E2	Broadleaved Deciduous Forest	20097
2nd Layer: LF, C, H	A3A10B2C1D1E2F2F5F7G2	Multi-Layered Broadleaved Deciduous Forest	20628
3rd Layer: LF, C, H	A3A10B2C1D1E2F2F5F7G2 F2F5F10G2	Multi-Layered Broadleaved Deciduous Forest With Emergents	20630

Concepts for Primarily Vegetated Areas

When observing plant communities and considering their growth forms, two factors are fundamental:

- **Physiognomy** , the overall appearance of the vegetation

- Vegetation Structure, which is defined as " the spatial" (Küchlerand Zonneveld, 1988). The structure describes the individual layers, usually characterized by **height and density** of the respective growth forms.



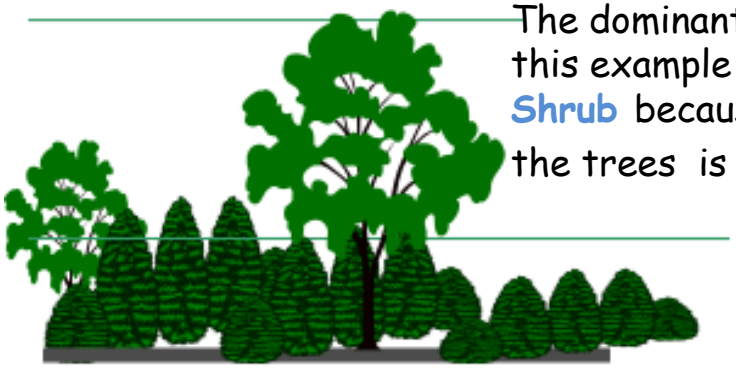
Life Form: Select Closed, Open or Sparse in combination with either Trees, Shrubs or Herbaceous

The dominant Life Form in this example is **Closed Herbaceous** because the cover of the trees is sparse.



The dominant Life Form in this example is **Open Shrubs**





The dominant Life Form in this example is **Closed Shrub** because the cover of the trees is sparse.

The dominant Life Form in this example is **Sparse Herbaceous.**



LAND COVER CLASSIFICATION SYSTEM

Description	No	Classes	Sub classes
- The USGS classification system (Anderson et al., 1976) is the most commonly used national classification system for remote sensing data. - For both satellite images and aerial photos - Based on land capability, vulnerability to management, and potential use	1	Urban or build-up land	Residential, commercial and service, industrial, transportation, communication and utilities mixed urban or built-up
	2	Agricultural	Cropland, pasture, orchards, vineyards, groves, confined feeding operations
	3	Rangeland	Herbaceous rangeland, shrub and brush rangeland, mixed rangeland
	4	Forest land	Deciduous forest land, evergreen forest land, mixed forest land
	5	Water	Streams and canals, lakes, reservoirs, bays and estuaries
	6	Wetland	Forested wetland, non-forested wetland
	7	Barren land	Dry salt flats, beaches, sandy areas other the beaches, bare exposed rock, strip mines, quarries, gravels, transitional areas, mixed barren land
	8	Tundra	Shrub and brush tundra, herbaceous tundra, bare ground tundra, wet tundra
	9	Perennial snow, ice	Perennial snowfields, glaciers

Global Land Cover system

GLC2000 Global Classes: Forest	
1	Tree Cover, broadleaved, evergreen **(LCCS >15% tree cover, tree height >3m)**
2	Tree Cover, broadleaved, deciduous, closed (or open to closed)
3	Tree Cover, broadleaved, deciduous, open **(LCCS 15% - 40% tree cover, tree height >3m)**
4	Tree Cover, needle-leaved, evergreen
5	Tree Cover, needle-leaved, deciduous
6	Tree Cover, mixed phenology or leaf type
7	Tree Cover, regularly flooded
8	Mosaic: Tree cover / Other natural vegetation

GLC2000 Global Classes: Non-Forest	
9	Shrub Cover, closed-open, evergreen (with or without sparse tree layer)
10	Shrub Cover, closed-open, deciduous (with or without sparse tree layer)
11	Herbaceous Cover, closed-open
12	Sparse Herbaceous or sparse shrub cover
13	Lichens & Mosses
14	Regularly flooded shrub and/or herbaceous cover
15	Cultivated and managed areas
16	Mosaic: Cropland / Tree Cover
17	Mosaic: Cropland / Other natural vegetation / Tree cover
18	Bare Areas
19	Water Bodies (natural & artificial)
20	Snow and Ice (natural & artificial)
21	Artificial surfaces and associated areas

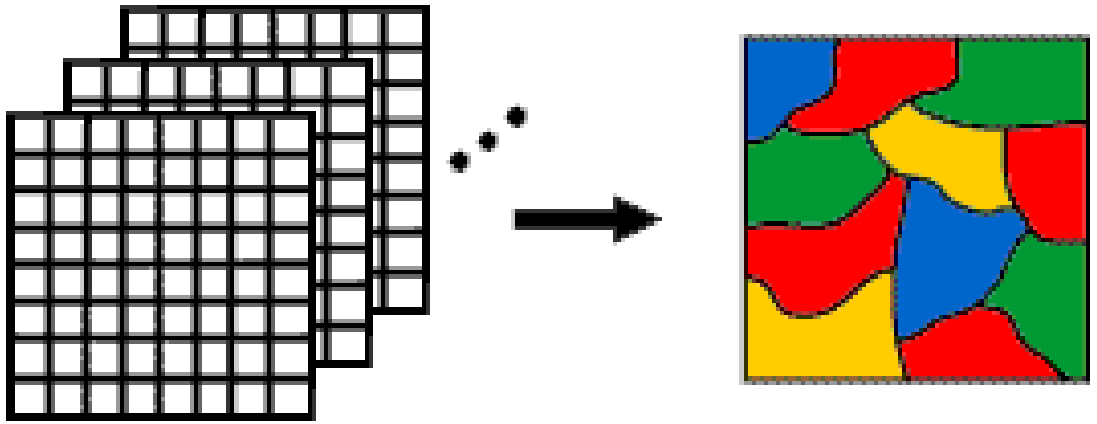
6.3 Image Classification

(Methods to make land cover maps from satellite images)

Why classify?

- Make sense of a landscape
  - Place landscape into categories (classes)
    - Forest, Agriculture, Water, etc
- Classification scheme = structure of classes = land cover/ land use classification system
  - Depends on needs of users
  - Depends on available satellite data

What is a Classified Image?



- Image has been processed to put each pixel into a category
- Result is a vegetation map, land use map, or other map grouping related features
- Categories are defined by the intended use of the map
- Can be few or many categories, depending on the purpose of the map and available resources
- *The resulting classified image will have errors!* Accuracy assessment is critical. Maps created by image classification should report an estimate of accuracy.

Basic Strategy: How do you do the classification?

- Use radiometric properties of remote sensor
- Different objects have different spectral signatures

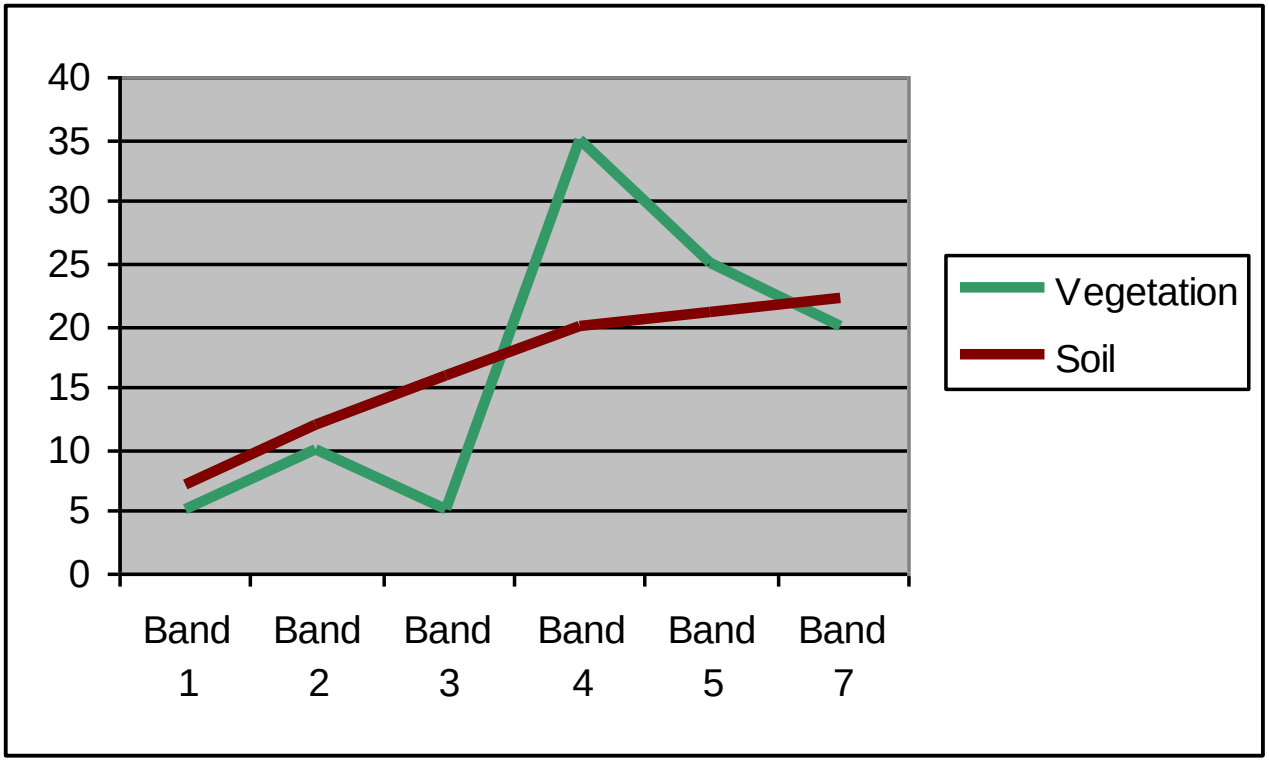
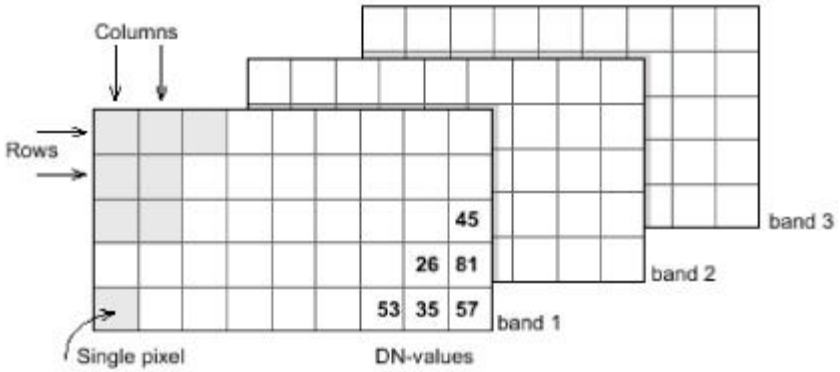
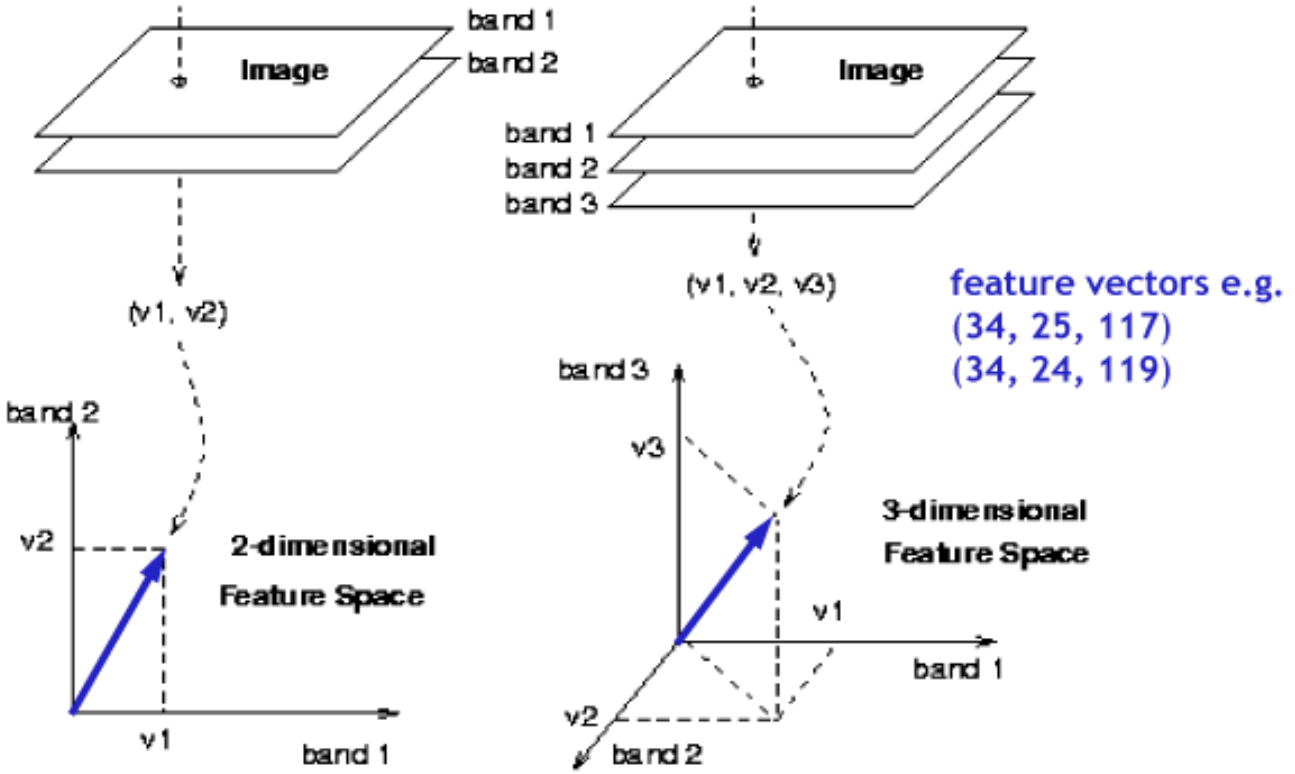


Image structure



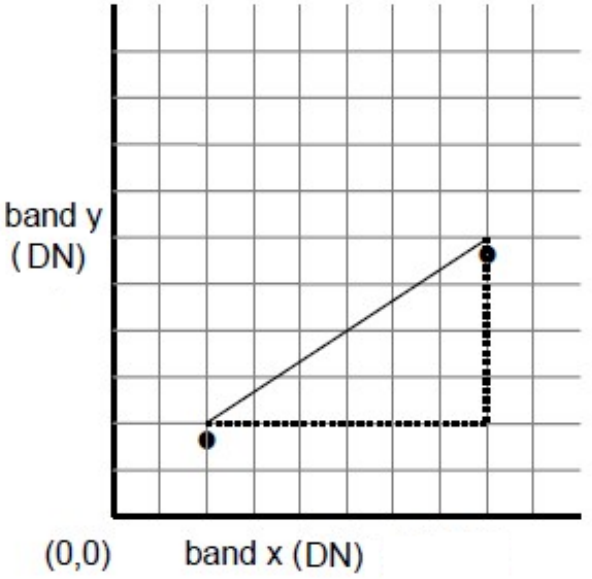
Multi-band image



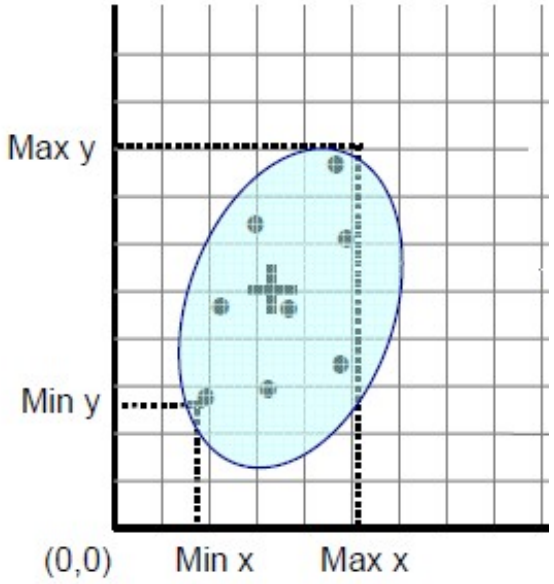
statistical pattern recognition

Concepts

- Euclidian distance: The magnitude difference between 2 pixels in a multi dimensional coordinate system generated by spectral bands (Unit: DN or radiant)
- Cluster: An area determined in the spectral coordinate system.



Euclidian distance



Cluster

Basis for 'classifying'

- method: pattern recognition
- use any/all of the following properties in an image to differentiate between land cover classes:-
  - spectral
  - spatial
  - temporal
  - directional
  - [time / distance-resolved (LIDAR)]

Spectral pattern recognition

- most widely used
- distinguish between different land cover classes from differences in the spectral reflectance (or more typically, image DN) in different wavebands

Spatial pattern recognition

- use spatial context to distinguish between different classes
  - e.g. measures of image texture, spatial context of 'objects ' derived from data.

Temporal pattern recognition

- the ability to distinguish based on spectral or spatial considerations may vary over the year
- use variations in image DN (or derived data) over time to distinguish between different cover types: e.g. variations in Tree over agricultural crops

Directional pattern recognition

- surface with different structures will tend to give different trends in reflectance as a function of view and illumination angles

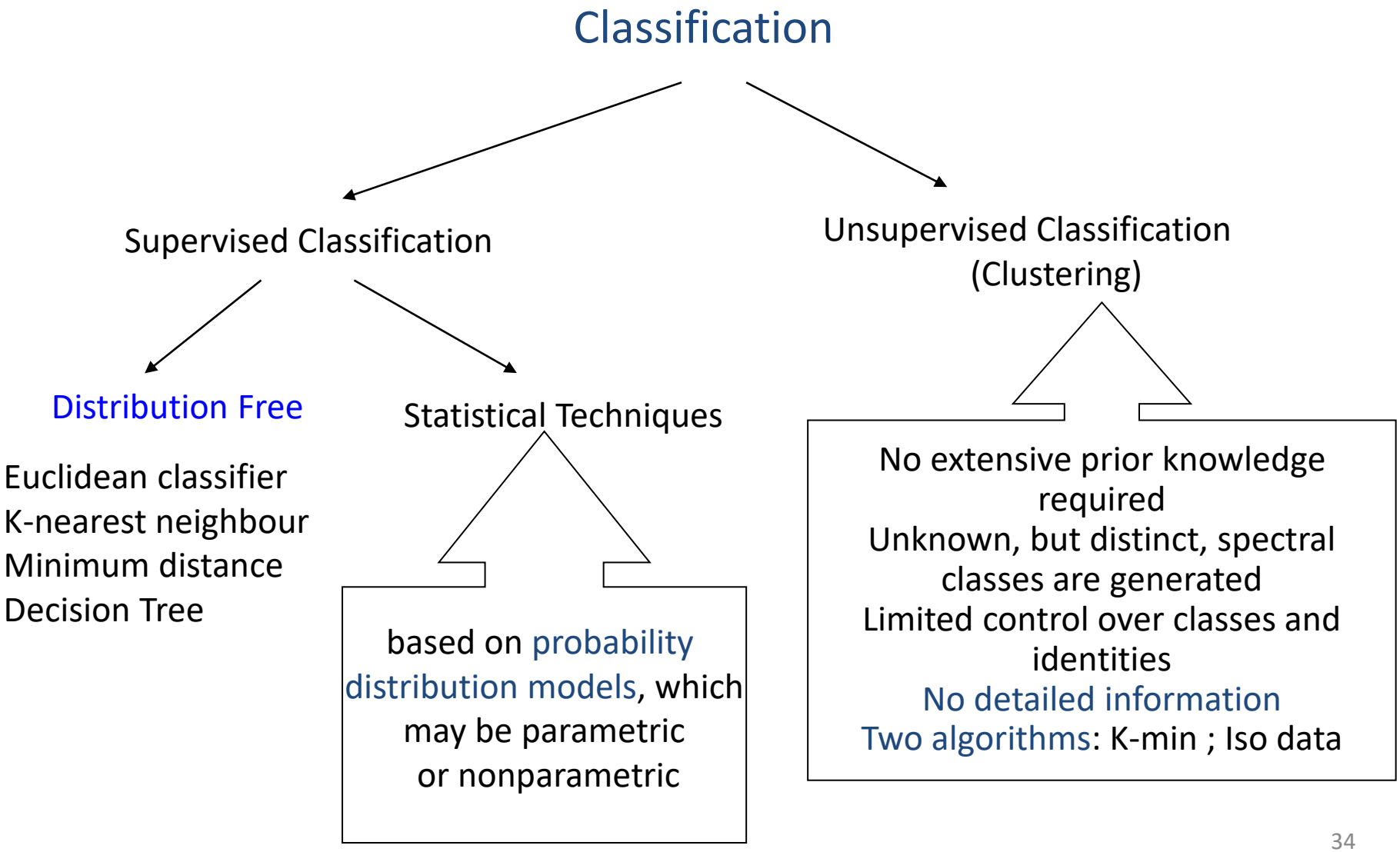
Land cover classification steps

1. Define why you want a classified image, how will it be used?
2. Decide if you really need a classified image
3. Define the study area
4. Select or develop a classification scheme (legend)
5. Select imagery
6. Prepare imagery for classification
7. Collect ancillary data
8. Choose classification method and classify
9. Adjust classification and assess accuracy

Approaches to Classification

- Digitalization
  - Recognize the objects and draw their boundaries by human skills
  - The accuracy depending on the implementer
  - Suitable for very high resolution and small area.
- We need some form of automated (rule-based) classification algorithm to allow us to distinguish one surface type from another
  - Supervised Classification
  - Unsupervised Classification

Image Classification



Supervised classification

- The process of using samples (training data) of known identities to classify pixels of unknown identities
- Training stage (significant user input/expertise)
- Identify areas of cover types of interest (map, ground survey, spectral characteristics) in bands of an image

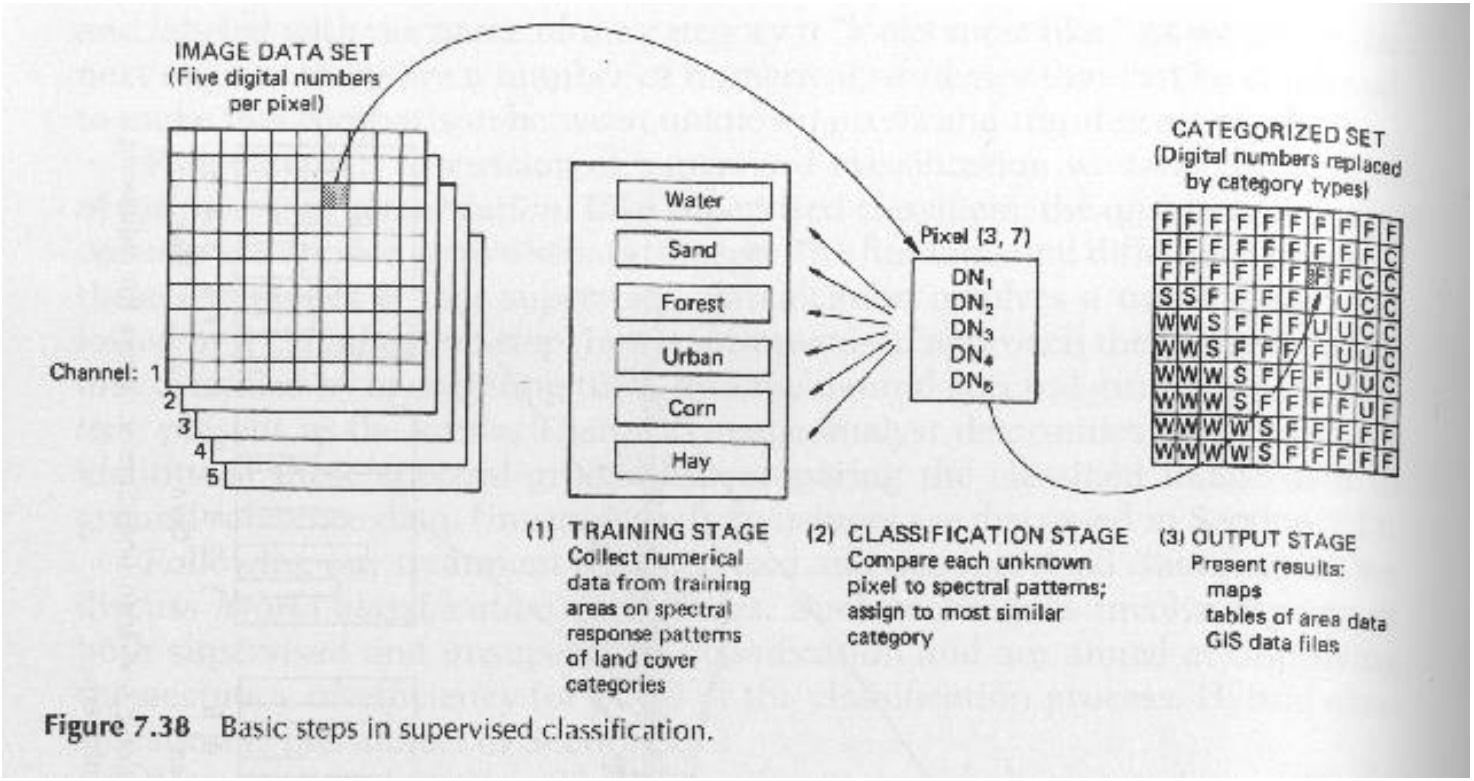


Figure 7.38 Basic steps in supervised classification.

From Lillesand, Kiefer and Chipman (2004)

Training regions

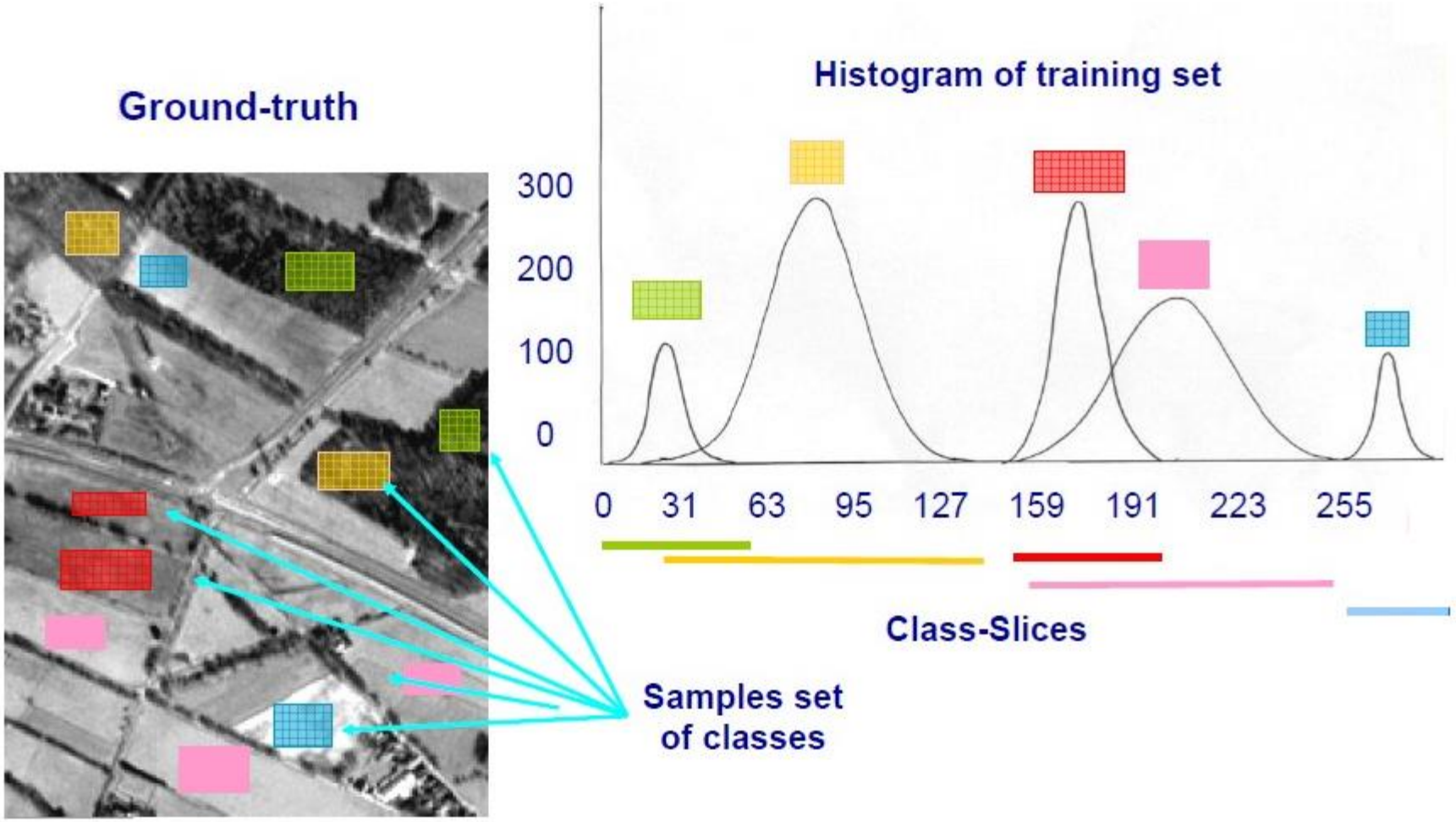
- **Training areas should be homogenous**. This can be tested by graphic histograms, numeric summaries, 2-band scatter plot for investigating separability of feature classes by pairs of bands, 3-D plot of 3-band feature space.

- **One large 'uniform' training area per feature class is preferable** to several smaller training areas.

- **Easy to determine an extract in the image:** a topographic map, nautical chart, or aerial photos to assist, though differential GPS observations may help.

- **The minimum size is critical**: Note CCRS statement for MSS: individual training area should be minimum of **3 - 4 pixels East-West by 6 pixels North-South**. [e.g. Swain and Davis, IDRISI] state (10 x # bands used), e.g. area of 40 pixels if all four MSS bands used (or approximate 6 pixels x 7 pixels).

Examples of training regions

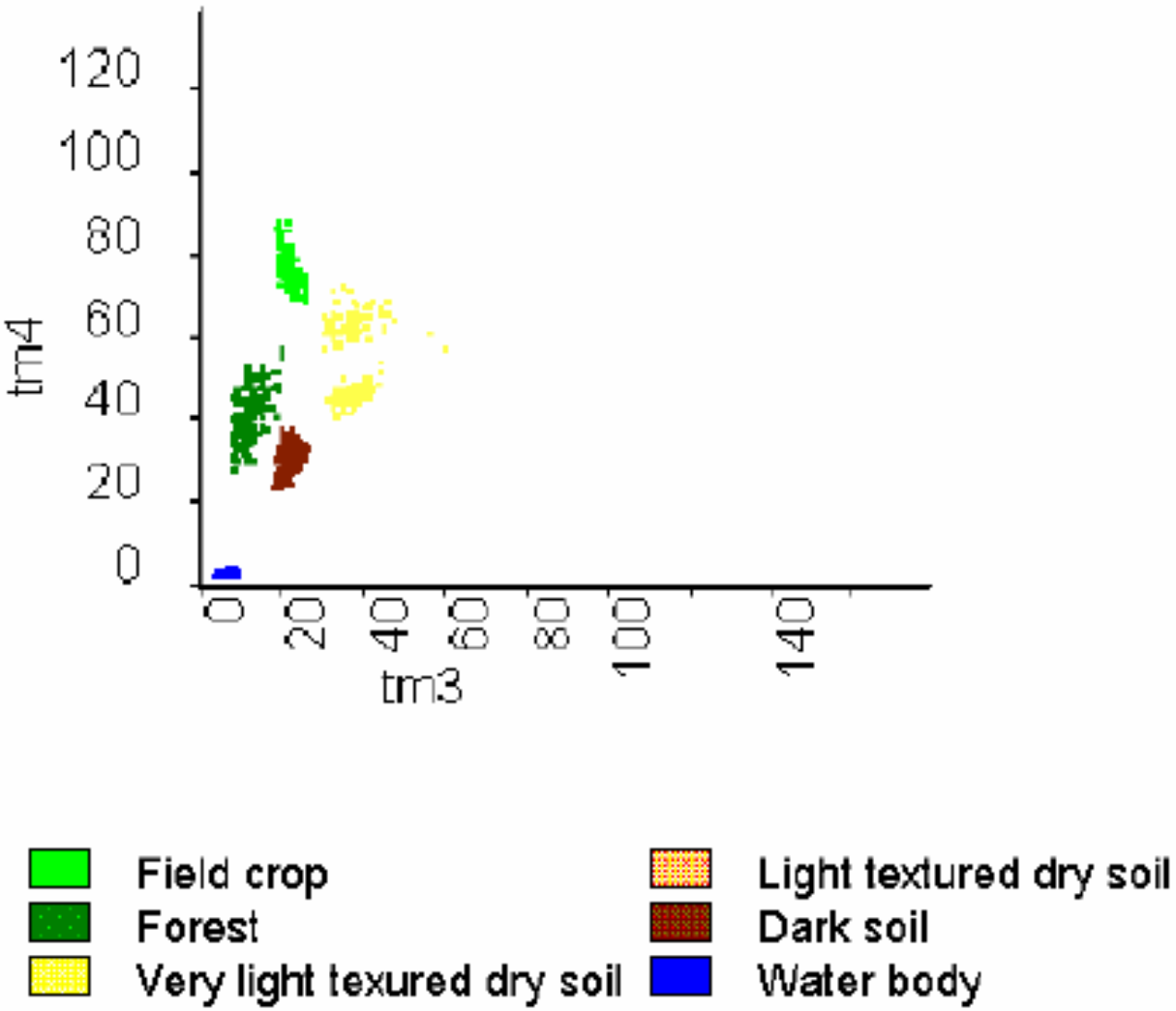


Spectral scatter plot of training regions

- Two dimensional plot (two spectral bands) of training region.

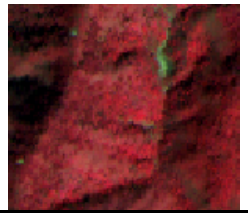
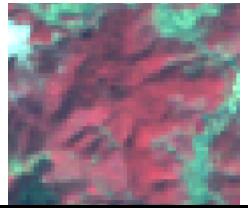
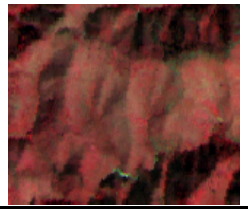
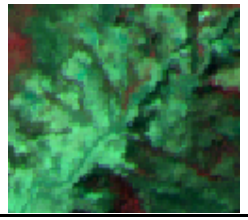
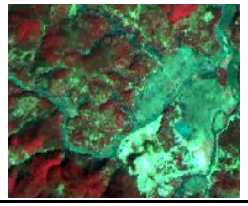
- The higher number of used bands, the higher dimensional of samples

- Each samples correspond to a certain cover type

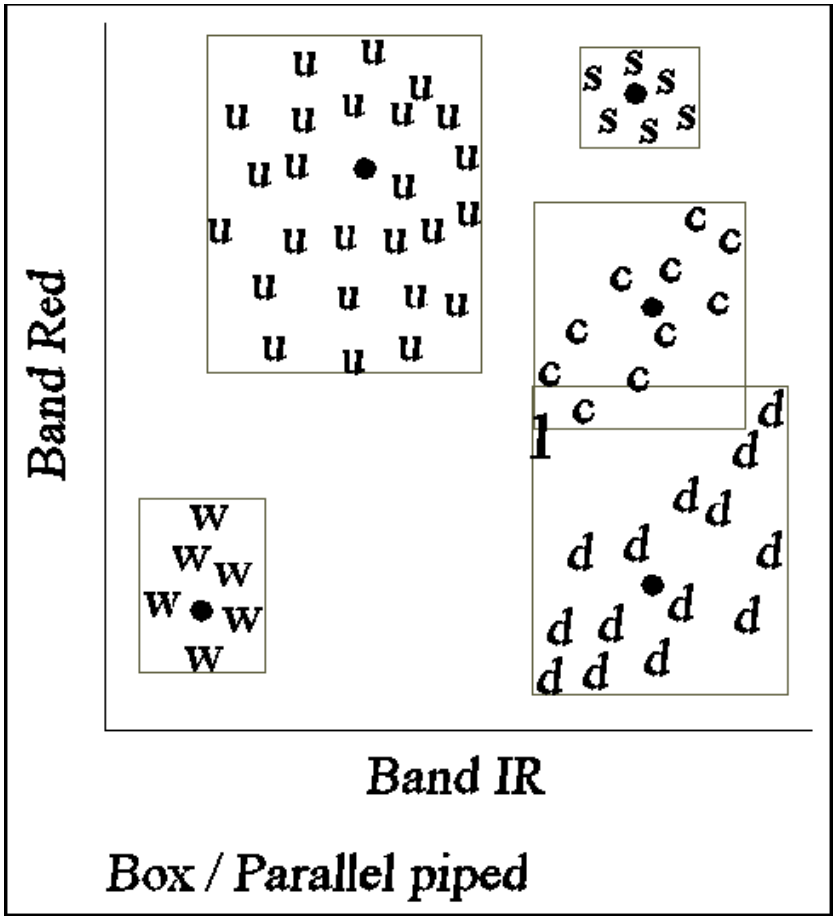


(UNEP-ITC RS/GIS, 2005)

Examples of training regions

Shrub land		- Light red - Uncertain shape	- Distributed in dow nhill area, buffer zone of the forest, - Often located near by bare land area or sometimes close to residential areas.	
Plantation forest		- Red color - Fine texture	- Located near by residential areas.	
Grass land		- Very light red or purple - Uncertain shape	- Often distribute in the dow nhill area	
Annual crop		It has bright or sea green depends on the grow ing period	-Distribute near by settlement	
Settlement		-Special texture -Bright or dot w hite, clear on image	-Distribute on flat terrain	

Supervised classification: Parallelepiped/ Box Approach

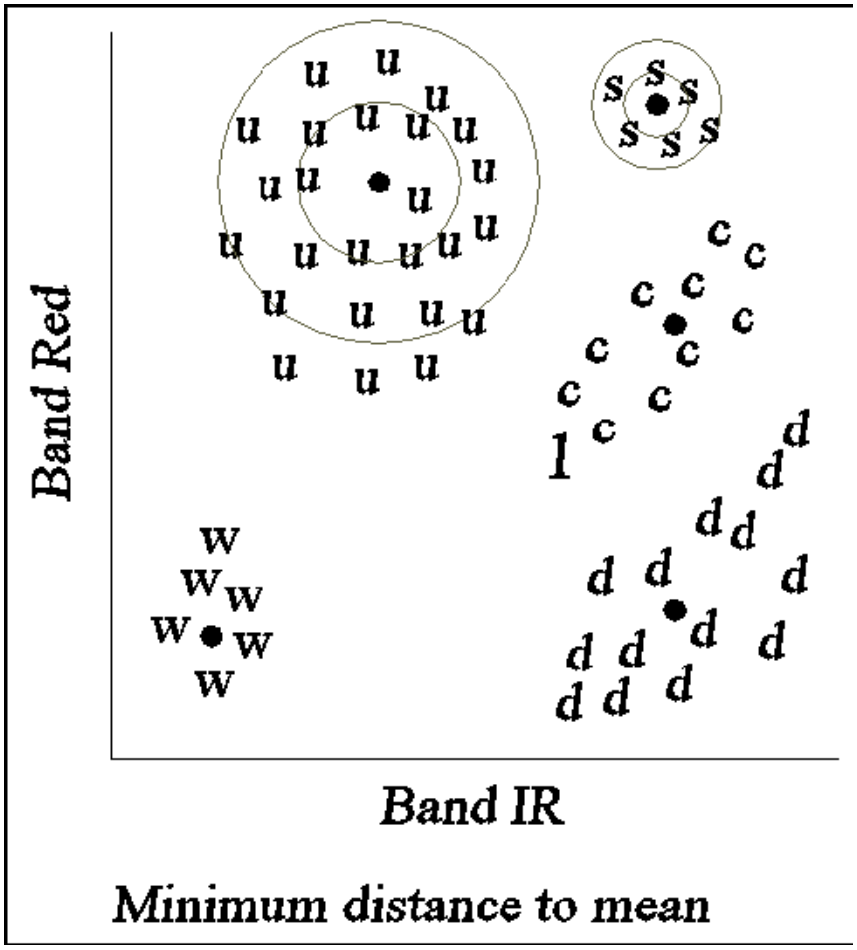


The Box classifier is the simplest classification method: In 2-D space, **rectangles are created** around the training feature vector for each class; in 3-Dimension they are actually boxes (blocks).

The position and sizes of the boxes can be exactly around the feature vectors (Min-Max method), or according to the **mean vector** (this will be at the center of a box) and the **standard deviations of the feature vector**, calculated separately per feature (this determines the size of the box in that dimension)

(UNEP-ITC RS/GIS, 2005)

Supervised classification: Minimum distance to means



(UNEP-ITC RS/GIS, 2005)

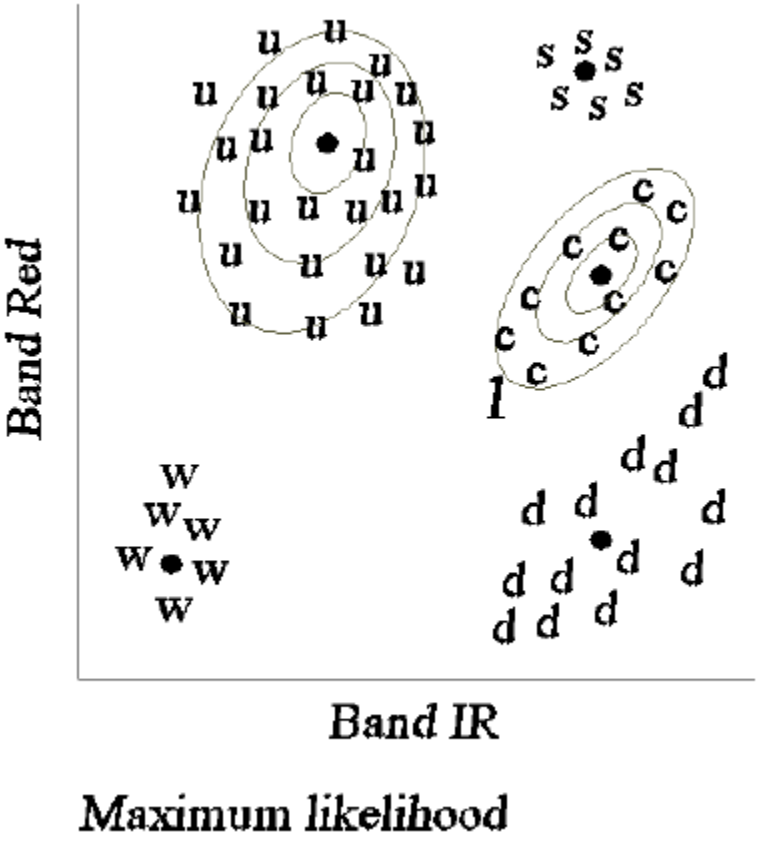
The Minimum Distance-to-mean classifier: first calculates for each class the **mean vector** of the training feature vectors.

Then, the feature space is partitioned by giving to each feature vector the class label of the nearest mean vector, according to **Euclidean metric**.

Usually it is possible to specify a maximum distance threshold:

If the nearest mean is still further away than that threshold, it is assumed that none of the classes is similar enough and the result will be **"unknown**".

Supervised classification: Maximum Likelihood



(UNEP-ITC RS/GIS, 2005)

Classifiers assume that the **feature vectors** of each class are (statistically) distributed according to a multivariate normal probability density function.

The training samples are used to estimate the parameters of the distributions. The boundaries between the different partitions in the feature space are placed where the decision changes from one class to another. They are called **decision boundaries**

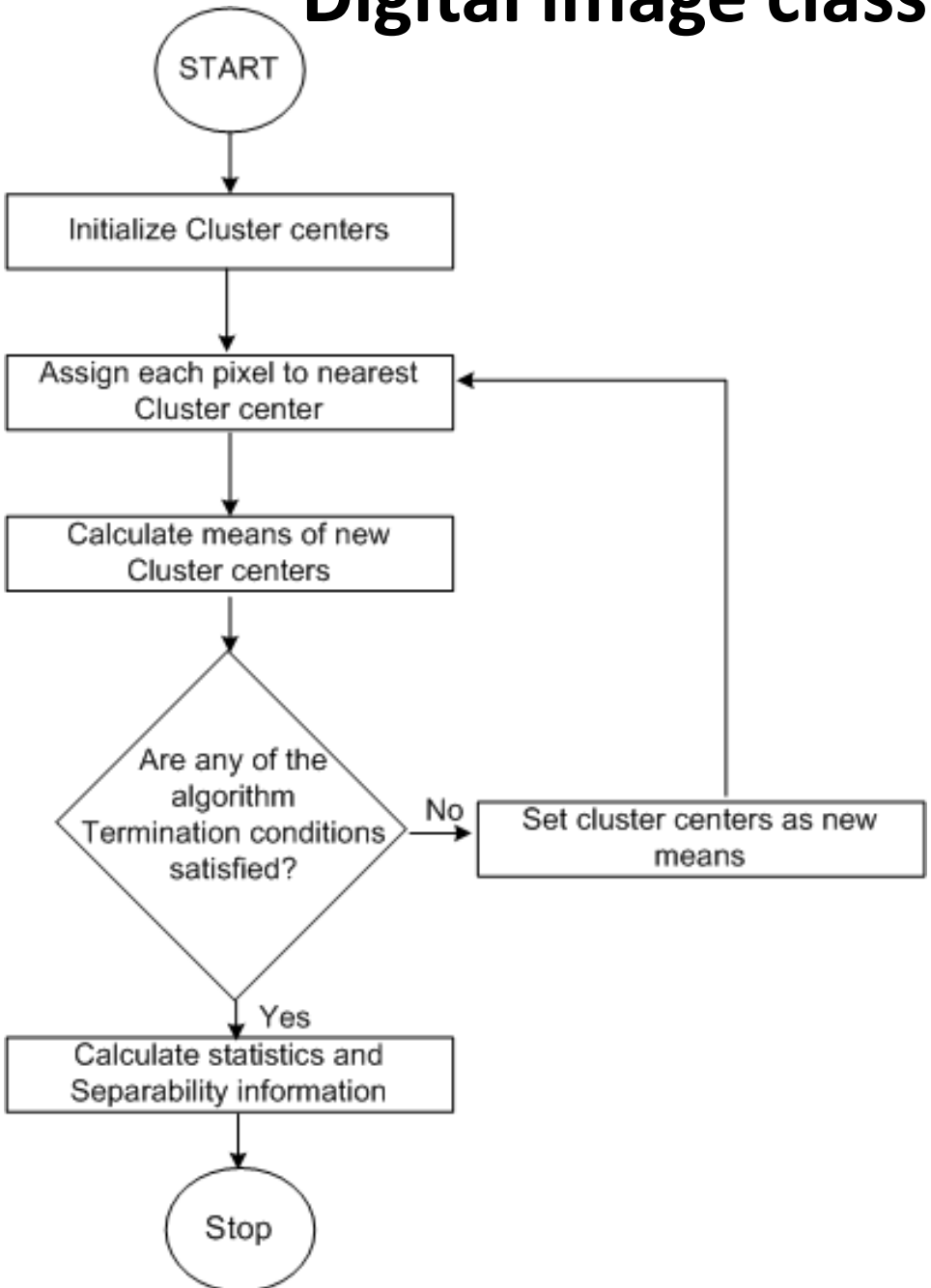
- Most sophisticated; achieves good separation of classes
- Requires strong training set to accurately describe mean and covariance structure of classes

Supervised classification

- It is tied to specific areas of known identity, called training areas.
- It is not necessary to match the spectral categories with the informational categories of interest.
- The operator may be able to detect serious errors in the classification by examining how training data have been classified

(UNEP-ITC RS/GIS, 2005)

Digital image classification: Unsupervised



K-means

- Accepts from analyst a number of clusters to be located in the data

- Each pixel is assigned to the cluster whose mean vector is closest

- After all pixels have been classified, revised mean vectors for each cluster are calculated. The revised means are used as basis to reclassify the image data.

- Procedure continues until no significant change in locations of class mean vectors

Widely used variant is ISODATA algorithm

Unsupervised Classification (clustering)

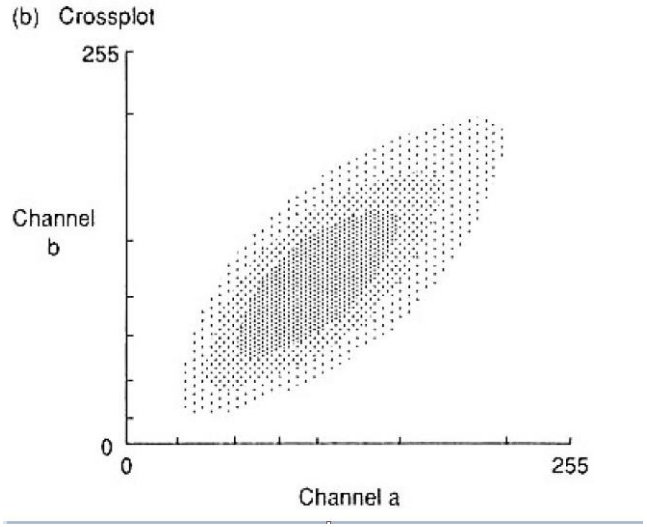
- Little input from user required (few assumptions)
  - BUT means results hard to interpret (may not represent classes we recognise)
- cluster pixels in feature space based on some measure of their proximity
- interpretation of results / assigned classes
  - can be useful, e.g. in picking up variations within what would otherwise be distinguished as a single class e.g. stressed/unstressed crop in a single field)
  - clusters can be of little intrinsic value in themselves
    - e.g. sunlit trees, shaded trees is perhaps not a useful discrimination if one simply wants to classify 'trees', and so clusters may have to be combined

Unsupervised Classification: K-means

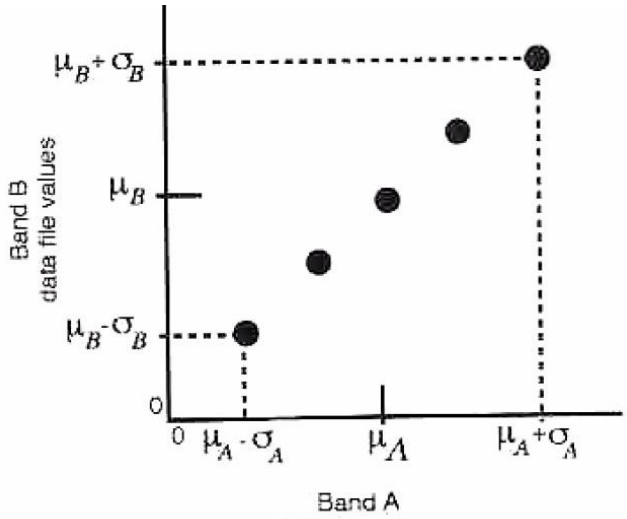
- A large number of clustering algorithms exist
- K-means
  - input number of clusters desired
  - algorithm typically initiated with arbitrarily-located 'seeds' for cluster means
  - each pixel then assigned to closest cluster mean
  - revised mean vectors are then computed for each cluster
  - repeat until some convergence criterion is met (e.g. cluster means don't move between iterations)
  - computationally-expensive because it is iterative

Unsupervised Classification: K-means

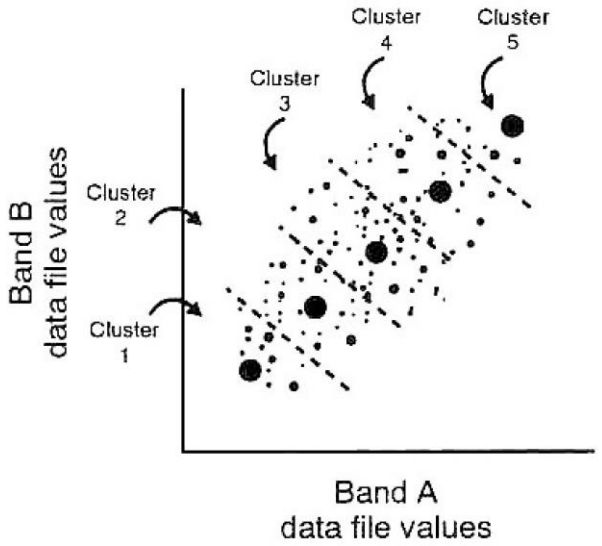
1/Spectral plot of whole image



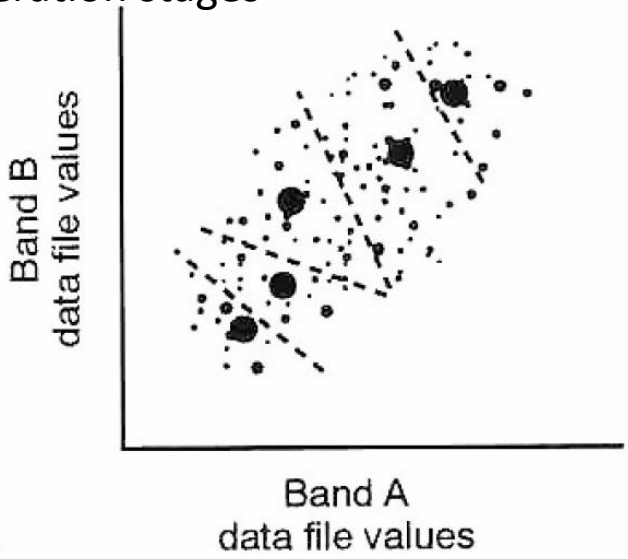
2/Input number of clusters desired (5 clusters)



3/ The fist separation



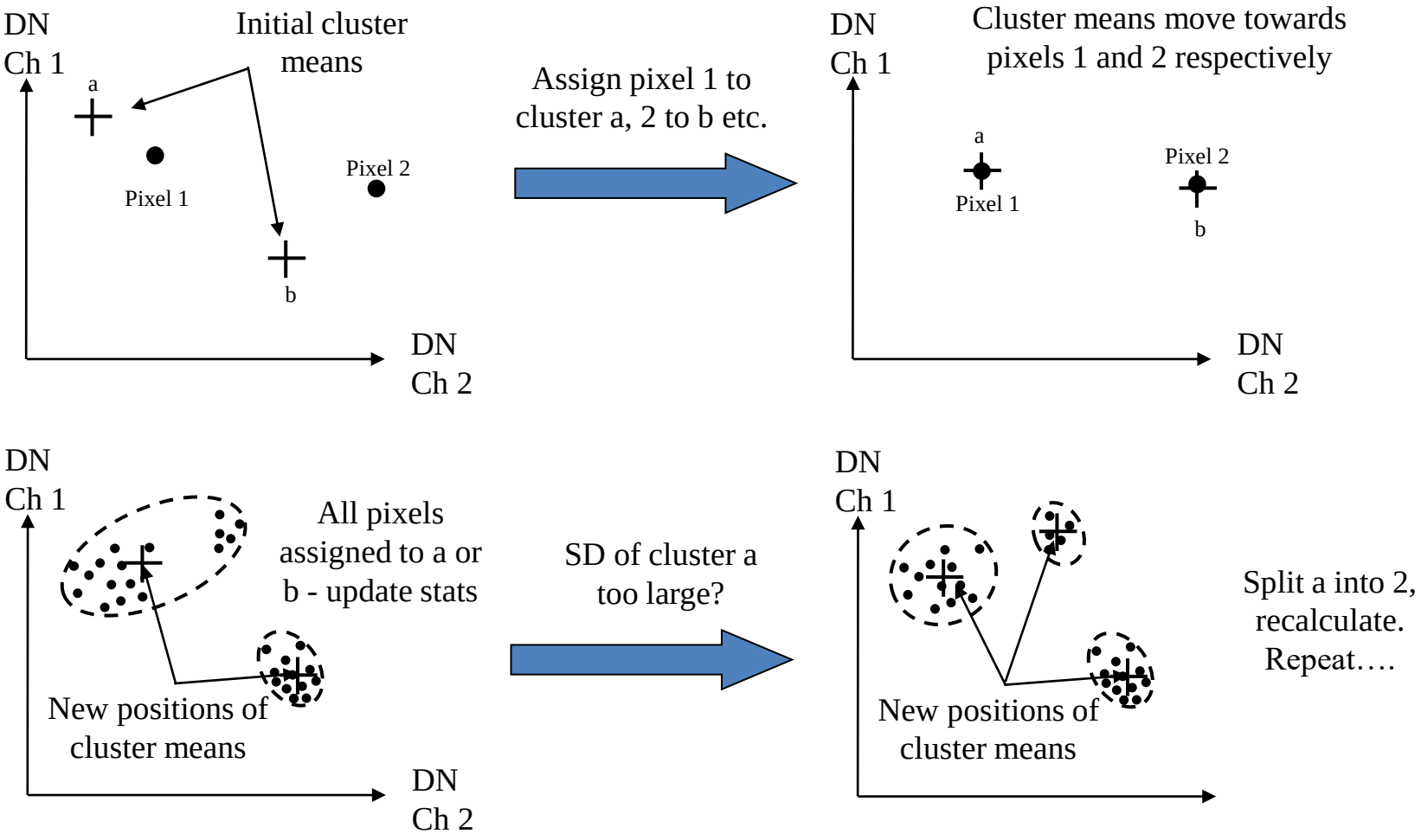
4/ Iteration stages



Unsupervised classification: ISODATA (Iterative self-organising data analysis) algorithm

- Same as K-means but vary number of clusters (by splitting / merging)
  - Start with (user-defined number) randomly located clusters
  - Assign each pixel to nearest cluster (mean spectral distance)
  - Re-calculate cluster means and standard deviations
  - If distance between two clusters < some threshold, merge them
  - If standard deviation in any one dimension > some threshold, split into two clusters
  - Delete clusters with small number of pixels
  - Re-assign pixels, re-calculate cluster statistics etc. until changes of clusters < some fixed threshold

ISODATA example: 2 classes, 2 bands



Unsupervised Classification

- Pros
  - No prior knowledge of the image area is required
  - Human error is minimized
  - Unique spectral classes are produced
  - Relatively fast and easy to perform
- Cons
  - Spectral classes do not represent features on the ground
  - Does not consider spatial relationships in the data
  - Can be very time consuming to interpret spectral classes
  - Spectral properties vary over time, across images

Post-classification filtering

- The result of a classification from RS data can often appear rather 'noisy'
- Can we aggregate information in some way?
- Simplest & most common way is majority filtering
  - a kernel is passed over the classification result and the class which occurs most commonly in the kernel is used
- May not always be appropriate; the particular method for spatial aggregation of categorical data of this sort depends on the particular application to which the data are to be put
  - e.g. successive aggregations will typically lose scattered data of a certain class, but keep tightly-clustered data

6.4 Accuracy assessments

Compare the classified (predicted) to an alternative (observed)

– Field data, manual interpretation of AP, etc

	Reference Data							
Classified Data	Deciduous Forest	Coniferous Forest	Wetland	Agriculture	Open Water	Urban	Total Pixels	User's Accuracy
Deciduous	90	5	4	1	0	0	100	90.00%
Coniferous	7	56	10	0	0	0	73	76.71%
Wetland	2	10	40	2	3	0	57	70.18%
Agriculture	5	1	0	90	0	1	97	92.78%
Water	0	0	2	0	80	0	82	97.56%
Urban	0	0	0	1	0	60	61	98.36%
Total Pixels	104	72	56	94	83	61	470	
Producer's Accuracy	86.54%	77.78%	71.43%	95.74%	96.39%	98.36%		88.51%

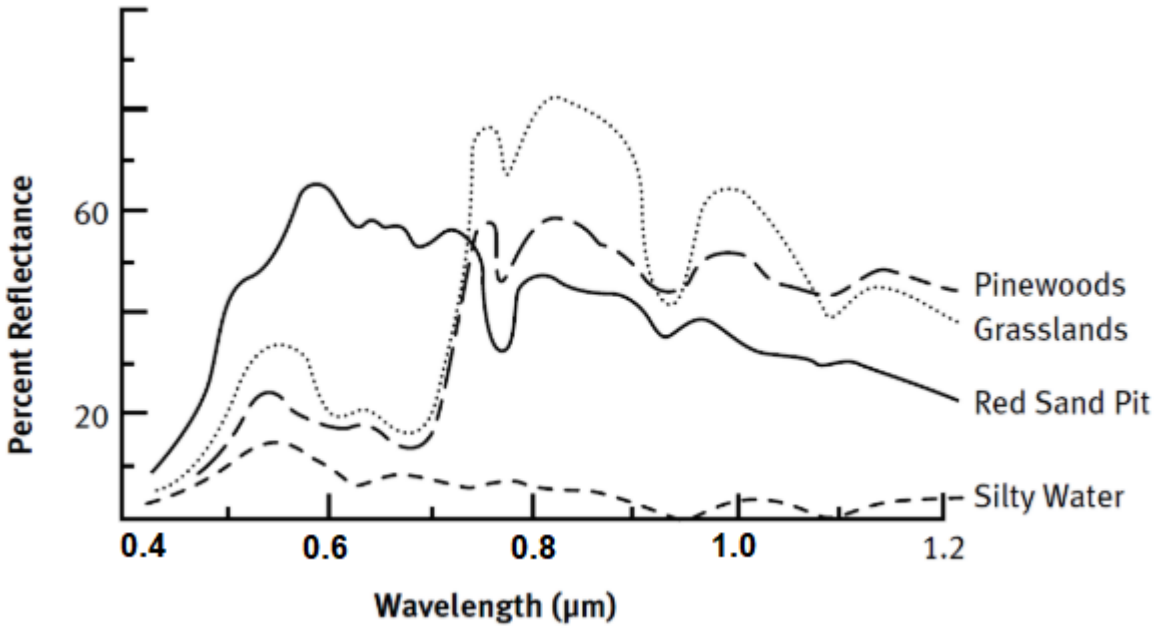
6.5 Extracting information from index images

- Normalized Difference Vegetation Index (NDVI)
- Enhanced Vegetation Index (EVI)
- Modified Soil-Adjusted Vegetation Index (MSAVI2)
- Soil Adjusted Total Vegetation Index (SATVI)
- Soil Adjusted Vegetation Index (SAVI)
- Soil-Adjusted Vegetation Index optimized for Agricultural Monitoring (OSAVI)
- Forest Canopy Density (FCD) methodology
- Advanced Vegetation Index (AVI)

- Normalized Difference Water Index (NDWI)
- Shadow Index (SI)
- Bare Soil Index (BI)
- Temperature Index (TI)

NDVI index

- Normalized Difference Vegetation Index



Spectral curves for various natural features (NASA, 2018)

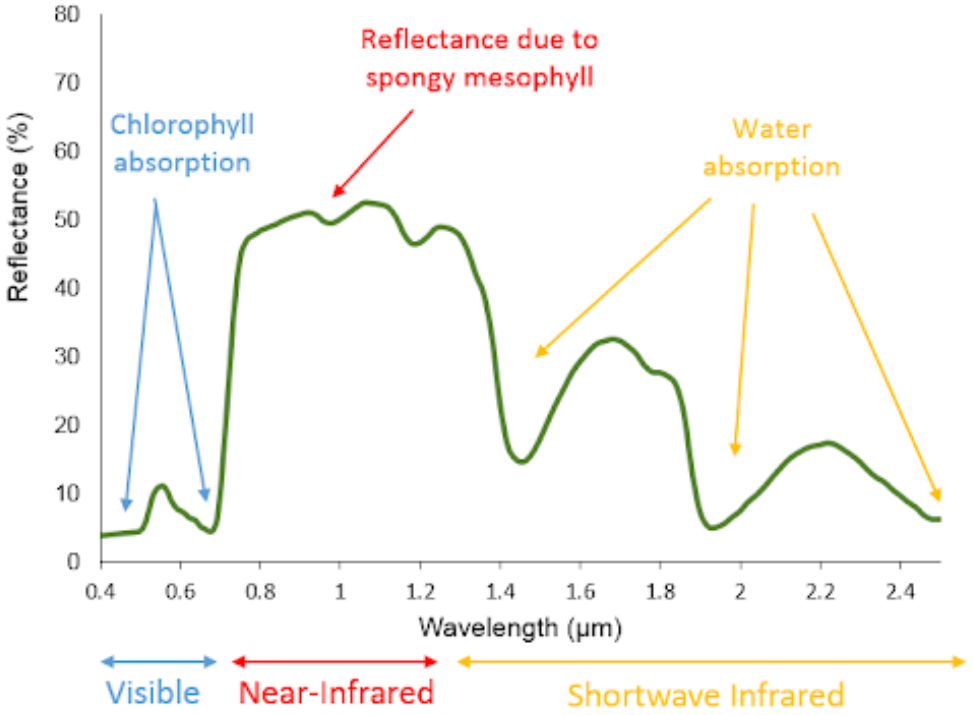
$$NDVI = \frac{(NIR - RED)}{(NIR + RED)}$$

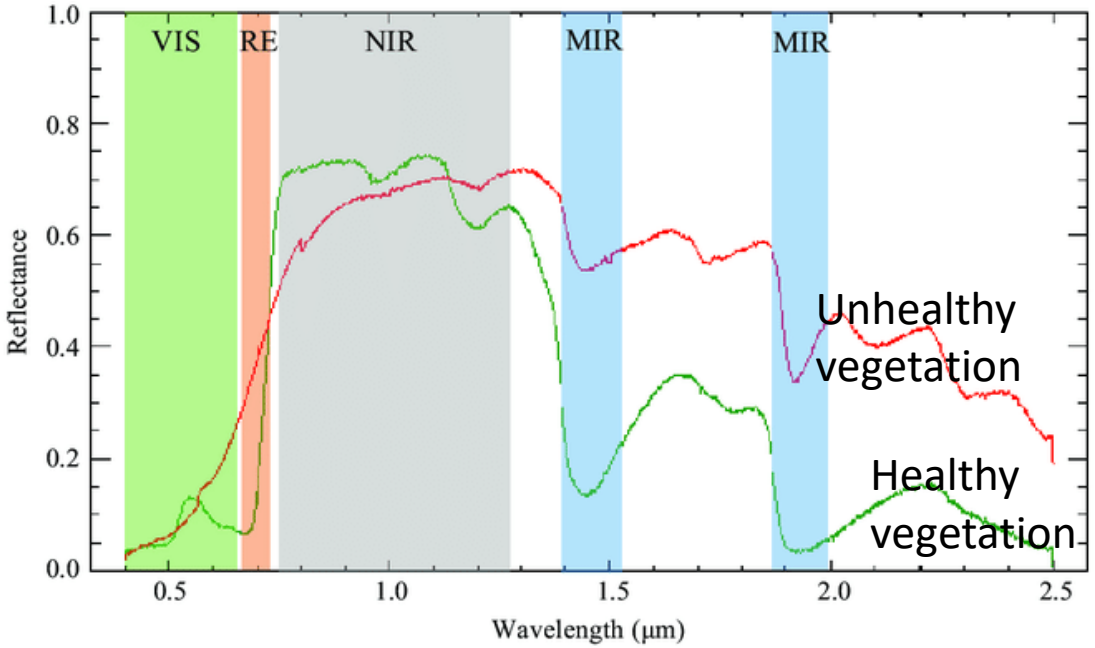
Leaves reflect near-infrared light (NIR) (plants use only visible light for photosynthesis).

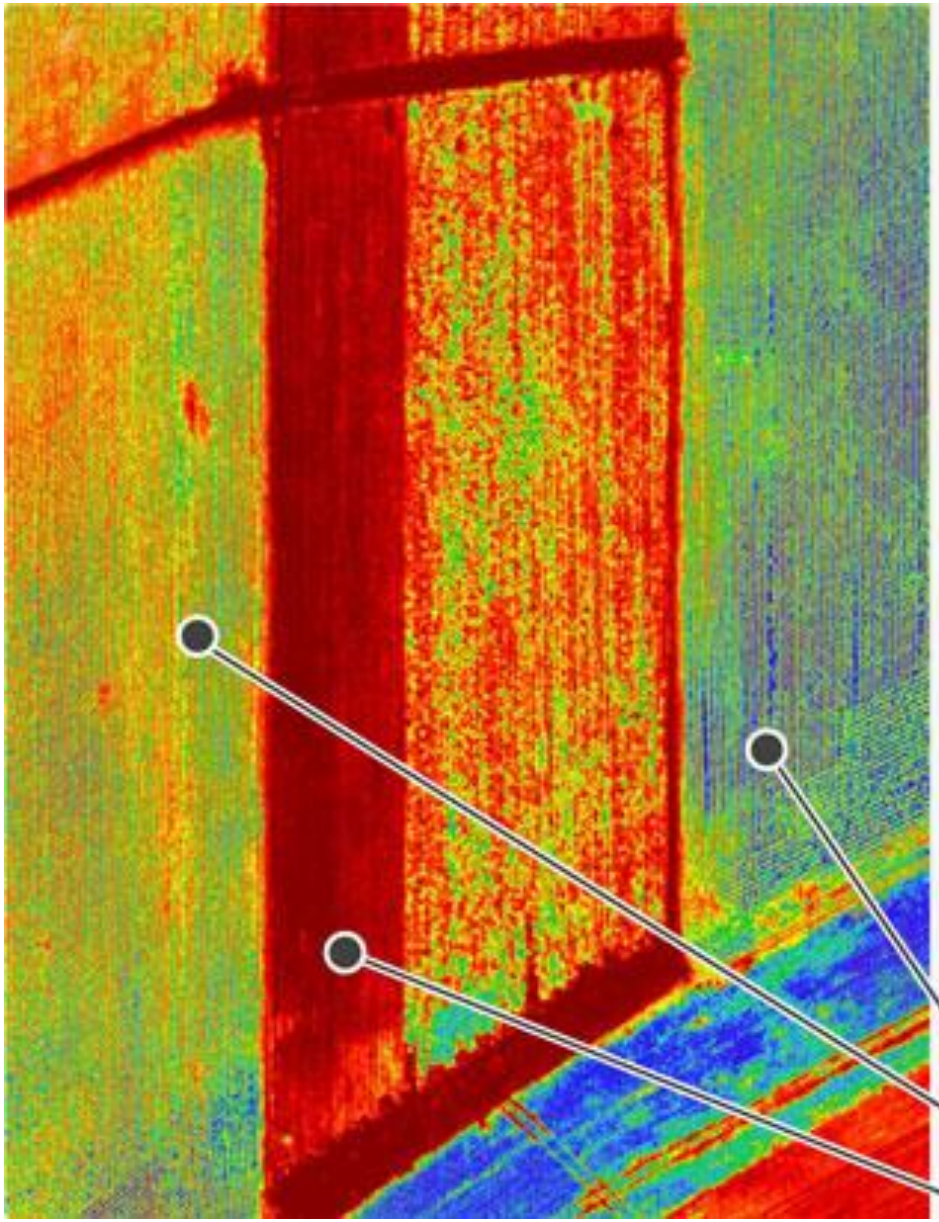
A healthy plant with good photosynthesis activity can be analyzed by comparing NIR with visible red light.

Unhealthy vegetation reflect more visible light and less NIR. Healthy vegetation will absorb most of the visible light falling on it.

$$NDVI = \frac{(NIR - RED)}{(NIR + RED)}$$






NDVI
0.96
0.86
0.76
0.66
0.56
0.47
0.37
0.27
0.17
0.07
Index range 0.33 - 1
Crops are feeling better or worse, but it is not clear from the NDVI map why and whether they are currently under stress. Any corrective action to improve their condition will not be reflected quickly in the NDVI picture and will be difficult to evaluate.
Index range 0 - 0.33
Crop has already died and it is impossible to tell from the picture how this happened and what intervention would help. Corrective intervention does not exist.
Healthy
Moderately Healthy
Dead crop

NDVI and Ecosystem

Ecosystem	Typical NDVI values	Location	References
Boreal forest	0.6-0.8	Alaska	Parent and Verbyla, 2010
Temperate forest	0.3-0.7	France	Pettorelli et al., 2006
Coastal rainforest	0.88-0.92	Solomon Islands	Garonna et al., 2009
Alpine pastures	0-0.35	Italy	Pettorelli et al., 2007
Annual grassland	0.15-0.45	California	Gamon et al., 1995
Desert	0.06-0.12	Sinai, Egypt	Dall'Olmo and Karnieli, 2002

Very low values of NDVI (<0.1) correspond to barren areas of rock, sand or snow.
Free standing water tend to be in the very low positive to negative values.

Modifications of NDVI

- Enhanced Vegetation Index (EVI) (Liu & Huete, 1995)

  A parameter to simultaneously correct soil and atmospheric effects

$$EVI = 2.5 * \frac{(NIR - RED)}{(NIR + C_1 * RED - C_2 * BLUE + L)}$$

  L corrects for soil background, C1 and C2 correct for aerosol scattering

  The standard EVI (NASA) in MODIS sensor: L=1, C1=6 and C2=7.5

- Modified Soil-Adjusted Vegetation Index (MSAVI2) (Xue et al., 2017)

$$MSAVI2 = \frac{(2 * NIR + 1 - \sqrt{(2 * NIR + 1)^2 - 8 * (NIR - RED)}}{2}$$

Analyze plant growth, desertification research, grassland yield estimation, LAI assessment, analysis of soil organic matter, drought monitoring, and the analysis of soil erosion. MSAVI2 is used for not completely covered with vegetation area and have exposed soil surface.

Modifications of NDVI

- **Soil Adjusted Vegetation Index (SAVI)/ Soil-Adjusted Vegetation Index optimized for Agricultural Monitoring (OSAVI)**

$$SAVI = \frac{(NIR - RED)}{(NIR + RED + L)} * (1 + L)$$

Default value: L=0.5

(Rondeaux et al., 1996)
Used for the vegetation density

$$OSAVI = \frac{(NIR - R)}{(NIR + R + 0.16)}$$

- **Soil-Adjusted Total Vegetation Index (SATVI)**

$$SATVI = \frac{SWIR1 - RED}{SWIR1 + RED + L} * (1 + L) - \frac{SWIR2}{2}$$

Measures both photosynthesizing and dead vegetation (Marsett et al., 2006; Qi et al., 2002)

Forest Canopy Density model (FCD)

- A tool to assess the regrowth of a forest canopy in logged-over tropical forests.
- Only uses the bands of the Landsat sensor
- Using data derived from four indices: Advanced Vegetation Index (AVI), Bare Soil Index (BI), Shadow Index or Scaled Shadow Index (SI, SSI) and Thermal Index (TI) (Rikimaru et al., 2002; Baynes, 2004)

- Advanced Vegetation Index (AVI)

$$AVI = \sqrt[3]{((B4 + 1) * (256 - B3) * (B4 - B3))}$$

- Shadow Index (SI)

$$SI = \sqrt{(256 - B2) * (256 - B3)}$$

- Bare Soil Index (BI)

$$BI = \frac{(B4 + B2) - B3}{(B4 + B2) + B3}$$

- Temperature Index (TI)

Surface temperature extracted from thermal band

Normalized Difference Water Index (NDWI)

- Measures the change in the water content of leaves by using the NIR and SWIR bands.
- NDWI is sensitive to the water content of plants as well as bodies of water
- Using for drought monitoring, recording yield reductions, reservoir discharge, lowering of groundwater levels.

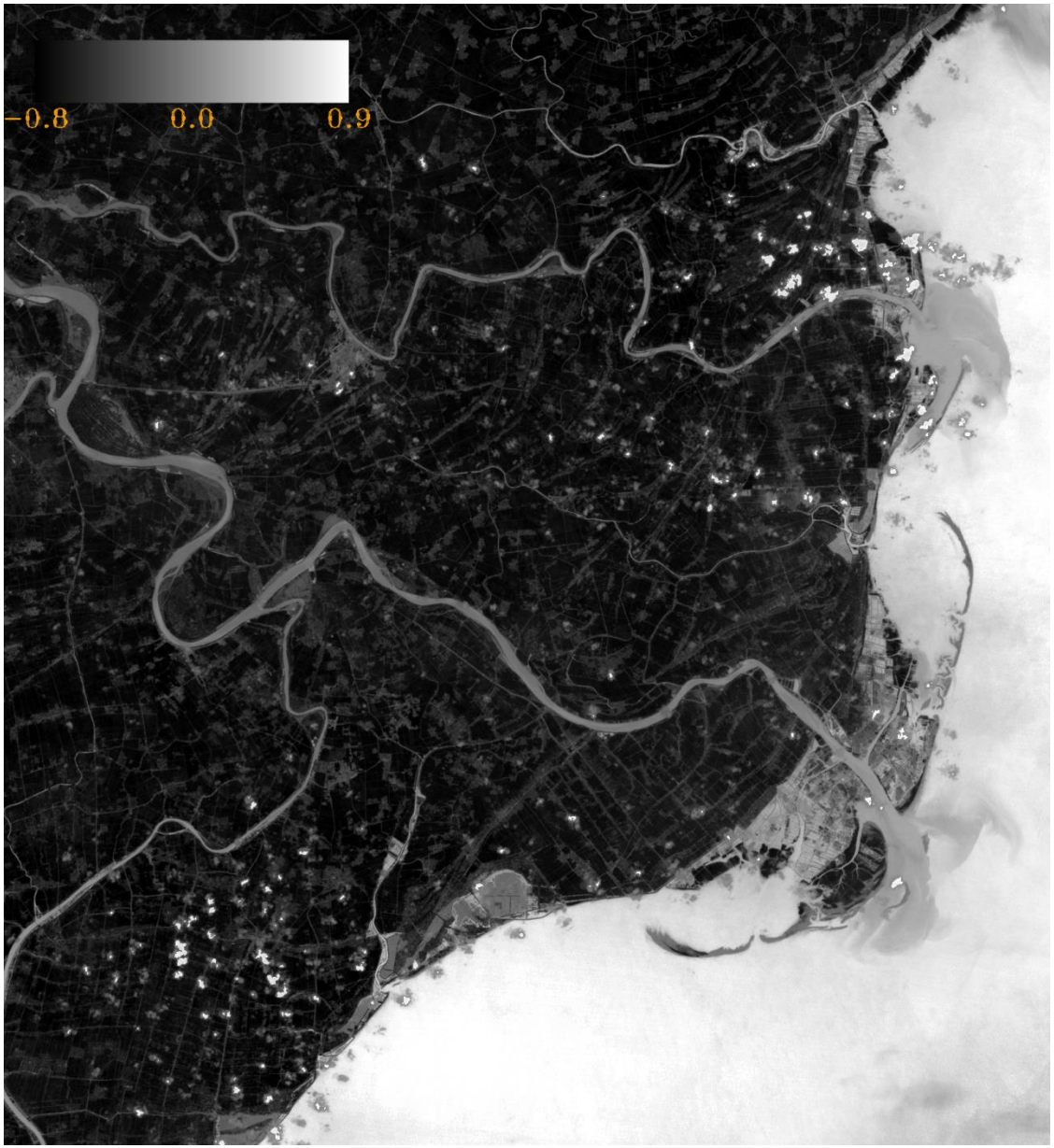
$$NDWI = \frac{(NIR - SWIR)}{(NIR + SWIR)}$$

- Or *NDWI = (Green-NIR)/(NIR+Green)*

NDVI_L720020831



NDWI_L720020831



NDWI = (Green-NIR)/(NIR+Green)

Thank you for your attention!